BOYD'S CONDUCTOR-11 MAHLER MEASURE CONJECTURE: PROOF OF THE SPLIT-INTEGRAL IDENTITY (C3), WITH AN EXACT STRUCTURAL ANALYSIS OF THE FAMILY S_k

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Abstract. Boyd's 1998 tables of conjectural identities $m(P) = r\,|L'(E,0)|$ between Mahler measures of two-variable polynomials and L-values of elliptic curves begin with the smallest possible conductor, $N = 11$. Two of the three conductor-11 identities were proved by Brunault via an explicit version of Beilinson's theorem on modular units. The third one, (C3), concerns the polynomial $S_0 = y^2 + (x^2+1)y + x^3$, which vanishes on the unit torus, and asserts that a *signed split integral* I_{split} of $\log|y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. We prove (C3). The proof identifies the split integral with a regulator integral along Samart's signed open chain $\tilde{\gamma}$; we close $\tilde{\gamma}$ by a small-branch compensating arc β_0 into a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$, and prove that its homology class is $2\gamma^-$, where γ^- generates $H_1(E,\mathbb{Z})^-$: the period ratio period$(C')/w_{\text{anti}}$ is a-priori an integer, and a ball-arithmetic (Arb) computation with certified error bounds pins it to 2 (in particular it is non-zero). Combined with the exact integral identity $\int_{\beta_0}\eta = \int_{\tilde{\gamma}}\eta$ and a direct regulator computation via Brunault's proved Siegel-unit formula—the symbol $\{x, y\}$ being a pair of modular units on $X_1(11)$, so no use of Bloch's diamond theorem is needed—this yields $I_{\text{split}} = b_{11}$; the sign is certified in interval arithmetic, and the identity agrees with the numerical value to 366 digits. For the family $S_k = y^2 + (x^2 + kx + 1)y + x^3$ we determine exactly the torus intersections (as a function of real k), the single case where x, y are modular units ($k = 0$), and the cases where the symbol is tempered with torsion-supported divisors; all further Boyd-type evaluations we find are numerical observations, recorded as conjectures ($k = 2, 3$ to 70 digits, $k = -4, -5, -6$ to 25 digits, PARI/GP cross-checked). At $k = -1$ (conductor 53) the closed-cycle/modular-unit mechanism provably fails; Samart's suggested analogue, for which no formula is on record, is left open. An appendix applies the same method to Samart's conductor-17 analogue $\tilde{n}(1) = b_{17}$, conditional on a normalization lemma in the conventions of Lalín–Ramamonjisoa. All computations are reproducible from the accompanying code; every certification step is carried out within interval arithmetic.

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1. Introduction

1.1. **Boyd's conductor-11 question.** Let $m(P)$ denote the logarithmic Mahler measure of a Laurent polynomial P. Boyd's systematic numerical experiments [5] produced dozens of conjectural identities of the shape

$$m(P_k) = r_k \, |L'(E_k, 0)|, \qquad r_k \in \mathbb{Q},$$

organised by the conductor N of the elliptic curve E_k cut out by $P_k = 0$. The smallest possible conductor of an elliptic curve over $\mathbb{Q}$ is $N = 11$ (there are no elliptic curves of conductor 1 through 10), so the conductor-11 rows head Boyd's table. Three polynomials of conductor 11 appear:

$$\text{(C1)} \qquad m\big((1+x)(1+y)(1+x+y)+xy\big) = 7\,b_{11},$$

$$\text{(C2)} \qquad m\big(y^2+(x^2+2x-1)y+x^3\big) = 5\,b_{11},$$

$$\text{(C3)} \qquad I_{\text{split}} = \pm\,b_{11},$$

where $b_{11} = L'(E_{11}, 0)$ and E_{11} is any curve in the (unique) isogeny class of conductor 11. Identities (C1) and (C2) were proved by Brunault [7] via his explicit version of Beilinson's regulator theorem for modular units on $X_1(11)$.

The third case is different in kind. The polynomial

$$S_0 = y^2 + (x^2+1)y + x^3$$

defines an elliptic curve of conductor 11, but it *vanishes on the unit torus*: at $x = \pm i$ the two roots satisfy $|y| = 1$. No rational relation between the Mahler measure itself and b_{11} is known:

$$m(S_0) = 0.40560295591501040\ldots,$$

and none was found by PSLQ searches with coefficient bound 10^8 (Boyd had reported the same negative finding). Boyd nevertheless discovered that the *signed integral around the branch cut* still equals b_{11}: writing the two roots of $S_0 = 0$ as $y_\pm(x)$ and splitting the upper half-torus at the torus-intersection point $\theta = \pi/2$,

$$\text{(C3)} \qquad I_{\text{split}} := \underbrace{\frac{1}{\pi}\int_0^{\pi/2} \log|y_-(e^{i\theta})|\,d\theta}_{I_1} - \underbrace{\frac{1}{\pi}\int_{\pi/2}^{\pi} \log|y_-(e^{i\theta})|\,d\theta}_{I_2} \stackrel{?}{=} \pm b_{11}.$$

Here y_- is the *continuous branch* of the smaller-modulus root on $|x| = 1$: at $\theta = 0$ the two roots coalesce at $y_-(1) = -1$, on $(0,\pi)\setminus\{\pi/2\}$ the branch is uniquely determined by $|y_-| < |y_+|$, and at the second node $\theta = \pi/2$ it is extended by continuity (at both nodes $|y_\pm| = 1$, so $\log|y_-| = 0$ there and the integrand is unaffected by the choice of value). Boyd's own words (quoted in [23]):

> "This is in accord with our contention that in case P vanishes on the torus, it is the integral of ω around a branch cut rather than $m(P)$, which should be rationally related to $L'(E,0)$."

Samart [23] records (C3) as an open identity in Boyd's "=?" conjectural notation (his eq. (4.1)). (Samart's eq. (4.1) writes the right-hand side as $-L'(E,0)$; the integral on the left, with Samart's own definition of y_-, is the positive number $+0.1521471\ldots$, so the sign there appears to be a typo — with our convention for y_- the identity reads $+b_{11}$, hence the $\pm$ in (C3).)

1.2. Main result.

Theorem (Theorem 4.9)**.** *$I_{\text{split}} = b_{11}$, with I_{split} the signed split integral of* (C3) *and $b_{11} = L'(E_{11}, 0)$.*

The proof combines exact algebra with computer-assisted steps. Everything outside the following list is exact (rational or symbolic arithmetic):

- the modulus ordering of Proposition 3.1 and the branch assignments of the closed chain, proved in ball arithmetic (§5.2);
- the homology class $\text{class}(C') = 2\gamma^-$, proved in ball arithmetic from the a-priori integer $\text{period}(C')/w_{\text{anti}}$ (§4.4, §5.2);
- the signs of the regulator integrals, proved by certified interval enclosures strictly contained in a half-line (§5.2);
- high-precision floating-point evaluations used only as consistency checks (e.g. the 366-digit agreement with b_{11}), never as proof steps.

Conventions: γ^- is the primitive anti-invariant generator of Lemma 4.2; $\eta(x, y) = \log|x|\, d\arg y - \log|y|\, d\arg x$; $\mathbb{Z}[E]^-$ is the quotient convention of [13]; D_E is the elliptic dilogarithm of §2, implemented verbatim from [13, Def. 5, eq. (10)].

1.3. Summary of results.

(1) **Proof of (C3).** We prove $I_{\text{split}} = b_{11}$ (Theorem 4.9). The proof identifies Boyd's split-integral chain with a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$ on E whose homology class is $2\gamma^-$, where γ^- generates $H_1(E, \mathbb{Z})^-$; the integrality argument is made rigorous by a ball-arithmetic (Arb) computation of the period ratio with certified error bounds (§5.2). The regulator side is closed by a direct computation: $\{x, y\}$ is a pair of modular units on $X_1(11)$, and Brunault's proved Siegel-unit regulator formula [8] evaluates $\int_{\gamma^-} \eta(x, y) = \pm 2\pi b_{11}$ through exact Siegel presentations and an exact weight-2 identity $F_{\text{total}} = -2 f_{11}$ (§4).

(2) **The conductor-17 analogue (appendix).** The same method is applied to Samart's conductor-17 conjecture $\tilde{n}(1) = b_{17}$ (Theorem A.1, Appendix A): the chain construction is verbatim parallel, the homology class is again certified in interval arithmetic, and the regulator side is closed by a published theorem of Lalín–Ramamonjisoa [13]; its status is conditional on the normalization lemma discussed there.

(3) **High-precision numerics.** (C3) is confirmed to 366 digits: $|I_{\text{split}} - b_{11}| = 9.26 \times 10^{-367}$, with b_{11} cross-checked against PARI/GP's `lfun` to 330 digits (previous public record: Boyd's 50 digits [6, p. 28]). (C1) and (C2) are independently reproduced to 52 digits.

(4) **Structural identity.** On $[0, \pi]$ one has $|y_-(e^{i\theta})| \le 1 \le |y_+(e^{i\theta})|$, hence the exact identity $I_1 + I_2 = -m(S_0)$, which reduces (C3) to the individual values $I_1 = (b_{11} - m(S_0))/2$, $I_2 = -(b_{11} + m(S_0))/2$ (Proposition 3.1).

(5) **Exact structure of the family S_k; numerical dichotomy.** For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ we determine exactly the torus intersections as a function of *real* k (Proposition 3.4: the torus meets the curve iff $k \in [-4, 2]$) and the modular-unit/temperedness structure (Proposition 3.5: x, y are modular units only at $k = 0$; at $k = 1$ the symbol is tempered with torsion-supported divisors, which suffices for the conductor-17 argument; at $k = -3, -2, -1, 2, 3$ the symbol is not cuspidal). The resulting dichotomy matches all numerical evidence: Boyd-type identities are proved for $k = 0$ and (conditionally on the normalization lemma) $k = 1$; observed numerically for $k \notin [-4, 2]$ in all computed cases (Conjecture 6.1); and no rational relation is found within the PSLQ bound 10^8 for $k = -1, -2, -3$.

(6) **New numerically confirmed identities** (70 digits, PARI/GP cross-checked): $m(S_2) = 2|L'(E_{37}, 0)|$, $m(S_3) = |L'(E_{79}, 0)|$, and the predicted-then-confirmed identities $m(S_{-4}) = \frac{7}{2}|L'(E_{37}, 0)|$, $m(S_{-5}) = \frac{1}{4}|L'(E_{359}, 0)|$, $m(S_{-6}) = \frac{1}{8}|L'(E_{997}, 0)|$.

(7) **Failure of the mechanism at conductor 53.** For $k = -1$ (curve `53.a1`) the mechanism that proves (C3) does not extend: there is no non-trivial anti-invariant closed cycle on the torus, and x, y are not modular units on the rank-1 curve (Proposition 3.8). Samart's remark mentioning a conductor-53 analogue has no formula on record and is left open.

1.4. **Organisation and reproducibility.** Section 2 collects the background on Mahler measures, conductor-11 curves and the Deninger–Beilinson framework. Section 3 proves the structural identity for S_0 and the exact structural results for the family S_k, and shows that the mechanism fails at conductor 53. Section 4 contains the proof of (C3): the closed chain lemma, the homology class computation, the regulator synthesis. Section 5 describes the numerical certification, including the ball-arithmetic proof of the period ratio (§5.2). Section 6 states the conjectures suggested by our computations. Appendix A applies the same method to the conductor-17 case $k = 1$; Appendix B lists the reproduction code. All scripts (Python/mpmath, PARI/GP, python-flint/Arb), raw outputs, and the sources of this paper are available in the accompanying repository.

2. Background

2.1. **Mahler measure.** For a non-zero Laurent polynomial $P \in \mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]$, the *logarithmic Mahler measure* is the average of $\log|P|$ over the unit torus $\mathbb{T}^n = \{|x_1| = \cdots = |x_n| = 1\}$:

$$m(P) = \int_0^1 \cdots \int_0^1 \log\left|P\left(e^{2\pi i t_1}, \ldots, e^{2\pi i t_n}\right)\right| dt_1 \cdots dt_n, \qquad M(P) = e^{m(P)}.$$

In one variable Jensen's formula gives, for $P(x) = a_0 \prod_{j=1}^{d}(x - \alpha_j)$,

$$m(P) = \log|a_0| + \sum_{j=1}^{d} \log^+|\alpha_j|, \qquad \log^+ v := \max(\log v, 0).$$

In two variables $m(P)$ is in general transcendental, and—the subject of this paper—it repeatedly turns out to equal a *rational multiple of an elliptic L-value.*

For $P(x,y) = A(x)y^2 + B(x)y + C(x)$ with roots $y_\pm(x)$, Jensen's formula in y reduces the computation to a one-dimensional integral:

$$m(P) = \frac{1}{2\pi}\int_0^{2\pi} \Big[\log|A(e^{i\theta})| + \log^+|y_+(e^{i\theta})| + \log^+|y_-(e^{i\theta})|\Big]\, d\theta. \tag{1}$$

All Mahler measures in this paper are computed from (1) with mpmath [12] at 60–300 digits.

The prototype of the whole story is Smyth's 1981 formula [24]

$$m(1 + x + y) = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2) = L'(\chi_{-3}, -1) = 0.3230659\ldots$$

Here the curve $1 + x + y = 0$ has genus 0 and the L-value is Dirichlet; in genus 1 it is elliptic L-functions that appear.

2.2. **Elliptic curves of conductor** 11**.** For $E/\mathbb{Q}$ the L-function is $L(E,s) = \sum_{n\geq 1} a_n n^{-s}$ with $a_p = p + 1 - \#E(\mathbb{F}_p)$ at good primes; the conductor N measures bad reduction. The minimal possible conductor is 11, and there is a single isogeny class (LMFDB [19] `11.a1--a3`), containing

$$X_1(11):\ y^2 + y = x^3 - x^2 \ (\texttt{11.a3}),$$
$$X_0(11):\ y^2 + y = x^3 - x^2 - 10x - 20 \ (\texttt{11.a1}).$$

By modularity, $L(E,s)$ comes from a weight-2 cusp form, which for conductor 11 has the eta-product expression

$$f_{11}(\tau) = \eta(\tau)^2\eta(11\tau)^2 = q\prod_{n\geq 1}(1 - q^n)^2(1 - q^{11n})^2, \qquad q = e^{2\pi i\tau}.$$

The completed L-function $\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(E,s)$ satisfies $\Lambda(E,s) = w\,\Lambda(E, 2 - s)$ with root number $w = +1$ here (rank 0). Since $\Gamma(s) = 1/s + O(1)$ and the functional equation forces $L(E,0) = 0$, one obtains the key constant identity

$$b_{11} := L'(E_{11}, 0) = \frac{11}{4\pi^2} L(E_{11}, 2) = 0.152147141725918049486227297478 63\ldots$$

Our script `code/b11.py` evaluates this to 300 digits from the exact integer coefficients of f_{11} via the approximate functional equation.

2.3. **The Deninger–Beilinson framework and the BMZ formula.** Deninger conjectured [10], and Rodríguez Villegas explained [21], that for *tempered* polynomials P (every face polynomial of the Newton polygon cyclotomic, equivalently the symbol $\{x, y\}$ taming trivially on $P = 0$) the Jensen-reduced integral can be rewritten as a *regulator integral*

$$\eta(x, y) = \log|x|\, d\arg y - \log|y|\, d\arg x$$

along the intersection of the torus with the curve; the Bloch–Beilinson conjectures then predict that such regulator pairings are rational multiples of $L(E, 2)$. Boyd's 1998 experiments [5] tabulated the conjectures; over 25 years most of them were proved (Rodríguez Villegas [21] in the CM conductors $27, 32, 36$, with lattice-sum reproofs by Rogers [17]; Mellit [16] and Brunault in conductor 14; Rogers–Zudilin [18, 22] in conductors $15, 20, 24$ (the conductor-24 formulas had first been obtained conditionally by Lalín–Rogers [15]); Lalín–Samart–Zudilin [14] in conductor 21; Brunault [7, 9] in conductor 11; see the survey [3]).

Brunault's thesis [7] made Beilinson's theorem on regulators of *modular units* (rational functions with divisors supported on cusps) fully explicit and applied it to $X_1(11)$, proving (C1) and (C2). Mellit–Brunault–Zudilin condensed this into the directly applicable *BMZ formula* [26]: for Siegel units $g_a(\tau) = q^{NB_2(a/N)/2} \prod_{n\equiv a}(1 - q^n) \prod_{n\equiv -a}(1 - q^n)$,

$$\int_{c/N}^{i\infty} \eta(g_a, g_b) = \frac{1}{4\pi} L(f_{a,b;c}, 2),$$

where $f_{a,b;c}$ is an explicit weight-2 modular form. Roughly: if x, y are modular units and the integration path joins cusps, the regulator integral is an L-value. This is the standard by which proof strategies are measured.

2.4. **Why (C3) is harder.** The (C1)–(C2) proof chain is: modular units + cusp-to-cusp path + BMZ. For (C3) the integration path ends at the torus-intersection point $P_{\pi/2} = (i, e^{i\pi/4})$, which is *not* a torsion point (proved rigorously by reduction at good primes, §5), so the naive boundary divisor is not cuspidal. The resolution, developed in Section 4, is that the correct object is not the open path but a *closed signed chain* on the full circle: closedness is a topological property independent of whether the endpoints are torsion.

3. The structural identity and the family S_k

3.1. **The structural identity for S_0.**

Proposition 3.1. *On $[0, \pi]$ one has $|y_-(e^{i\theta})| \le 1 \le |y_+(e^{i\theta})|$ (with $\max|y_-| = 1$ attained only at $\theta = 0, \pi/2$). Since $|y_+y_-| = |x^3| = 1$ on the torus,*

$$m(S_0) = \frac{1}{\pi}\int_0^\pi \log|y_+|\, d\theta = -\frac{1}{\pi}\int_0^\pi \log|y_-|\, d\theta = -(I_1 + I_2).$$

Proof. The modulus ordering on $[0,\pi]$ is proved by interval arithmetic: adaptive bisection with ball arithmetic certifies that $\log|y_-|$ is negative on $(0,\pi)$ except at the fold $\theta=\pi/2$ and the endpoint $\theta=0$, where it vanishes (`code/branch_certify.py`, §5; this proposition is computer-assisted in the sense of §5.2). Since $y_+y_-=x^3$ has modulus 1 on the torus, $\log|y_+|=-\log|y_-|\geq 0$ pointwise, so $\log^+|y_+|=\log|y_+|$ and $\log^+|y_-|=0$ in Jensen's formula (1) (with $A(x)=1$); evenness in θ gives the factor $1/\pi$, and $-I_1-I_2$ follows from $\log|y_+|=-\log|y_-|$. □

Corollary 3.2. *Conjecture* (C3) *is equivalent to the individual values*

$$I_1=\frac{b_{11}-m(S_0)}{2},\qquad I_2=-\frac{b_{11}+m(S_0)}{2}.$$

Remark 3.3. PSLQ searches on $(m(S_0),b_{11})$ with coefficient bound 10^8, and on $\{m(S_0),b_{11},\log 2,\log 3,\mathrm{Catalan},m(1+x+y)\}$ with bound 10^{10}, return no relation, supporting the expectation that $m(S_0)$ itself requires elliptic dilogarithms rather than elementary constants.

3.2. **The family S_k: exact structure and numerical observations.** Consider

$$S_k=y^2+(x^2+kx+1)y+x^3,$$

with associated elliptic curve E_k.

Proposition 3.4 (Torus intersections, exact)**.** *For real k, write $x=e^{i\theta}$, $\theta\in[-\pi,\pi]$. A point (x,y) of the torus $|x|=|y|=1$ lies on $S_k=0$ if and only if θ falls in one of the following two cases:*

(i) $\theta=0$ *and* $|k+2|\leq 2$*;*
(ii) $k+2\cos\theta=0$ *(solvable iff* $|k|\leq 2$*).*

Consequently:

(1) *the torus meets the curve if and only if* $k\in[-4,2]$*;*
(2) *for* $-2\leq k\leq 2$ *the* fold points $\theta=\pm\arccos(-k/2)$ *occur, where* $B=x^2+kx+1=0$ *and* $y^2=-x^3$ *has both roots of modulus* 1*;*
(3) *for* $-4\leq k\leq 0$ *an additional* branch-exchange intersection *at* $\theta=0$ *occurs, where* $S_k(1,y)=y^2+(k+2)y+1$ *has both roots on the unit circle — distinct for* $-4<k<0$*, a double root (tangency) at* $k=-4$ *(*$y=1$*) and at* $k=0$ *(*$y=-1$*);*
(4) *at* $k=2$ *the folds coalesce with the boundary* $\theta=\pi$*, where* $S_2(-1,y)=y^2-1$*.*

This matches Samart's observation $K\cap\mathbb{R}=[-4,2]$ *for the torus-intersection parameter interval* [23]*.*

Proof. Put $y=e^{i\phi}$. Since $x+x^{-1}=2\cos\theta$, one has $B=x^2+kx+1=x(k+2\cos\theta)$. Dividing $e^{2i\phi}+Be^{i\phi}+e^{3i\theta}=0$ by $e^{i\phi}$ and using $e^{i\phi}+e^{i(3\theta-\phi)}=2\cos\psi\, e^{3i\theta/2}$ with $\psi:=\phi-3\theta/2\in\mathbb{R}$ gives

$$2\cos\psi\, e^{i\theta/2}=-(k+2\cos\theta),\tag{2}$$

a real number. Comparing imaginary and real parts,

$$\cos\psi\,\sin(\theta/2)=0,\qquad 2\cos\psi\,\cos(\theta/2)=-(k+2\cos\theta).$$

The first equation gives two cases. (i) $\sin(\theta/2)=0$, i.e. $\theta=0$ on $[-\pi,\pi]$: the second equation reads $2\cos\psi=-(k+2)$, solvable iff $|k+2|\le 2$, and then $y=e^{i\phi}$ with $\cos\phi=-(k+2)/2$ is indeed a root of $S_k(1,y)=y^2+(k+2)y+1$. (ii) $\cos\psi=0$: the second equation forces $k+2\cos\theta=0$, solvable iff $|k|\le 2$; conversely, if $k+2\cos\theta=0$ then $B=0$ and $S_k(e^{i\theta},y)=y^2+e^{3i\theta}$ has the two roots $y=\pm e^{i(3\theta+\pi)/2}$, both of modulus 1 — these are the fold points, with $\psi=\pm\pi/2$ realising (2). The union of the two parameter ranges is $[-4,0]\cup[-2,2]=[-4,2]$, giving (1); items (2)–(4) are read off directly (at $k=2$ the fold $\arccos(-1)=\pi$; at $k=-2$ the fold $\arccos(1)=0$ coincides with the case-(i) point). ☐

The conductors, confirmed by PARI [20] `ellfromeqn` and `ellglobalred`, are

$$k=-3,-2,-1,0,1,2,3\ \longmapsto\ N=83,\ 91=7\cdot 13,\ 53,\ 11,\ 17,\ 37,\ 79.$$

Denote by $\tilde n(k)$ the modified Mahler measure along the signed chain (when no fold points exist, $\tilde n(k)=m(S_k)$). The numerical observations (70 digits; PARI/GP cross-checks), with their evidence labels, are summarised in Table 1.

k	N	root number	fold c/π	result	status (digits)
-3	83	-1	none	ratio $0.8529175\ldots$	none found (PSLQ)
-2	91	-1	none	ratio $0.6339454\ldots$	none found (PSLQ)
-1	53	-1	1/3	ratio $0.7392026\ldots$	mech. fails (Prop. 3.8)
0	11	$+1$	1/2	$\tilde n=-b_{11}$	**proved**, (C3)
1	17	$+1$	2/3	$\tilde n=+b_{17}$	**proved** (cond.), Thm. A.1
2	37	-1	none	$m=\tilde n=2\,b_{37}$	numerical, 70 digits
3	79	-1	none	$m=\tilde n=b_{79}$	numerical, 70 digits
-4	37	-1	none (tangent)	$m=\frac{7}{2}\,b_{37}$	numerical, pred., 25 digits
-5	359	-1	none	$m=\frac{1}{4}\,b_{359}$	numerical, pred., 25 digits
-6	997	-1	none	$m=\frac{1}{8}\,b_{997}$	numerical, pred., 25 digits

TABLE 1. The family S_k: conductors and Boyd-type evaluations, with evidence labels. "Numerical" rows are conjectural identities (Conjecture 6.1); "none found" means that PSLQ searches with coefficient bound 10^8 found no rational relation. Here $b_N:=|L'(E_k,0)|\ge 0$ and $\tilde n(k)$ is Samart's signed split-integral; for $k=0$ one has $\tilde n(0)=-I_{\text{split}}$ in the notation of (C3), so the proved identity appears with a minus sign.

The torsion data (PARI `elltors`/`ellorder`) reveal the true dichotomy (Table 2):

Proposition 3.5 (Modular units in the family, exact)**.**

k	N	torsion	ord$(0,0)$	Boyd-type identity
0	11	$\mathbb{Z}/5\mathbb{Z}$	5	proved (modular units, (C3))
1	17	$\mathbb{Z}/4\mathbb{Z}$	4	proved ($\tilde{n} = b_{17}$, Thm. A.1)
$-3, -2, -1, 2, 3$	$83, 91, 53, 37, 79$	trivial	∞	not modular units (Prop. 3.5)

TABLE 2. Torsion versus Boyd-type identities (exact data).

(1) *For $k = 0$ ($E = X_1(11)$, $E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/5\mathbb{Z}$) the divisors of x, y are supported on rational torsion points, and these are exactly the rational cusps under the modular identification (§4), so x, y are modular units; a Boyd-type identity is* proved *in this case* (C3). *For $k = 1$ ($E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/4\mathbb{Z}$) the divisors of x, y are supported on rational torsion points and the symbol $\{x, y\}$ is tempered (its Newton face polynomials are cyclotomic, Appendix A), so $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$; this is the only K-theoretic input used by the conductor-17 proof (Theorem A.1). We do* not *assert that x, y are modular units for $k = 1$: on the natural modular model $X_0(17)$, which has only two cusps, a divisor supported on all four rational torsion points cannot be cuspidal, and no other modular identification is provided.*

(2) *For $k \in \{-3, -2, -1, 2, 3\}$ the rational torsion of E_k is trivial (PARI `elltors`, exact). The point $(0, 0) \neq O$ is then non-torsion, and since each E_k is the strong Weil curve of its conductor, the cusps of $X_0(N)$ map to torsion points of E_k (Manin–Drinfeld), hence to O; a modular unit on E_k would have divisor supported on $\{O\}$ alone, while $x = 0$ meets S_k at $(0, 0)$ and $(0, -1)$. Hence x, y are* not *modular units and the modular-unit regulator mechanism does not apply to them. For $k = -1$ the closed-cycle ingredient fails as well (Proposition 3.8).*

Remark 3.6 (The structural dichotomy, as numerical evidence). Together with Proposition 3.4 this yields a clean *observed* dichotomy (Tables 1 and 2): for integer $k \notin (-4, 2)$ (no genuine torus intersection) the classical Deninger mechanism is expected to apply directly, and identities $m(S_k) = r_k \, |L'(E_k, 0)|$ with small rational r_k are observed numerically (Conjecture 6.1); for integer k with $-4 < k < 2$ the only case where x, y are modular units is $k = 0$, while $k = 1$ has a tempered symbol with torsion-supported divisors (but no modular-unit structure, Proposition 3.5), and for $k = -1, -2, -3$ (trivial torsion, branch-exchange intersections at $\theta = 0$) no rational relation is found within the PSLQ bound 10^8. We record this as evidence for Conjecture 6.1, not as a theorem.

Remark 3.7. The $k = 1$ row ($\tilde{n}(1) = b_{17}$) was first found as an independent high-precision confirmation of the conductor-17 analogue stated by Samart; it is proved as Theorem A.1 (Appendix A).

3.3. **The conductor-53 case: failure of the mechanism.** Samart [23, §4] mentions a conductor-53 analogue (for $k = -1$) in a single sentence, with no formula, precision, or reference. Since the exact statement to be tested is not on record, we cannot adjudicate it directly; what we can and do test is whether the *mechanism* of this paper — a closed anti-invariant torus cycle paired against a modular-unit symbol — extends to $k = -1$. It does not, for three independent reasons.

Proposition 3.8 (Failure of the closed-cycle/modular-unit mechanism at $k = -1$)**.** *For S_{-1} (curve* `53.a1`*):*

(1) Topological obstruction (exact). *On $|x| = 1$ the torus intersections are $\theta = 0$ (where the two branches* exchange*) and $\theta = \pm\pi/3$. The space of closed anti-invariant chains with breakpoints at the intersections and big/small branch choices is $\mathbb{Z} \cdot (1, 1, 1, 1)$: the boundary matrix of the anti-invariant symmetrization is a $\pm 1/0$ integer matrix whose kernel over $\mathbb{Q}$ is exactly one-dimensional, computed by exact rational linear algebra (`code/k53_smith.py`), and the generator $(1, 1, 1, 1)$ has period 0 — in fact exactly, not numerically: $(1, 1, 1, 1)$ is the sum of the two sheets over the full circle, and the two branches have $u = 2y + B = \pm\sqrt{B^2 - 4x^3}$, so $1/u_+ + 1/u_- \equiv 0$ pointwise and $\sum_{\text{branches}} dx/u$ vanishes identically (the continuous-root loop, which closes only after* two *turns, vanishes by the same cancellation). Hence the natural chain space contains* no non-trivial closed anti-invariant torus cycle*: the first ingredient of the (C3) mechanism does not exist at $k = -1$.*

(2) Algebraic obstruction. *The functions x, y are not modular units. Indeed*

$$\operatorname{div}(x) = [P] + [-P] - 2[O], \qquad \operatorname{div}(y) = 3[P] - 3[O],$$

with $P = (0, 0)$ a Mordell–Weil generator*:* `53.a1` *has trivial rational torsion and rank 1 (PARI* `ellorder`*$(P) = 0$,* `ellidentify`*). The curve is the strong Weil curve of conductor 53; the two cusps of $X_0(53)$ map to torsion points of E (Manin–Drinfeld), hence to O since $E(\mathbb{Q})_{\text{tors}}$ is trivial, so a modular unit on E would have divisor supported on $\{O\}$ alone — while the support of* $\operatorname{div}(x)$ *and* $\operatorname{div}(y)$ *contains the non-torsion point P. The Beilinson–Brunault mechanism therefore has no symbol to apply to.*

(3) Numerical observation. *For the natural candidate (the half-loop L_1, an open chain), the period ratio $0.5492906\ldots$ with w_{anti} shows no integrality at the computed precision, and the regulator integral over $2\pi b_{53}$ equals $0.7392026\ldots$; PSLQ searches with coefficient bound 10^8 find no rational relation.*

None of this excludes identities arrived at by other means; it rules out the specific mechanism that proves (C3)*. In the absence of a precise proposed formula, we leave the conductor-53 case open.*

4. Proof of (C3)

Throughout, $E : S_0 = 0$ with invariant differential $\omega = dx/u$, $u = 2y + x^2 + 1$ (quartic model $u^2 = x^4 - 4x^3 + 2x^2 + 1$; the model constant $\kappa = 1$ is proved exactly in §5).

4.1. **Modular units.** The quartic model's invariants give $j = -2^{12}/11 = j(X_1(11))$. Exact group-law computation (`code/torsion.py`, exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$, not numerical) on the quartic model, with its point at infinity as neutral element, for $A = (0, 0)$ on S_0 (the point $(x, u) = (0, 1)$ in quartic coordinates):

$$2A = (0, -1), \qquad 4A = (1, 0) = -A \implies 5A = O.$$

The S_0-model's projective closure has *two* points at infinity, $O_1 = [0 : 1 : 0]$ (the image of the quartic neutral element) and $Q_\infty = [1 : -1 : 0]$ (which the birational identification with $X_1(11)$ below takes to the origin O); on this model

$$\operatorname{div}(x) = [(0,0)]+[(0,-1)]-[O_1]-[Q_\infty], \qquad \operatorname{div}(y) = 3[(0,0)]-2[O_1]-[Q_\infty]$$

are supported on 5-torsion points. These are exactly the rational cusps of $X_1(11)$: the birational identification of $S_0 = 0$ with $E : y^2 + y = x^3 - x^2 = X_1(11)$ is exact (Riemann–Roch computation composed with PARI's exact minimal transform, round-trip verified, `code/kappa_exact.py`), and under the standard modular isomorphism (infinite cusp $\mapsto O$, $\omega_f = dx/(2y + 1)$) the rational cusps P_v, $v \in (\mathbb{Z}/11\mathbb{Z})^\times/\pm 1$, map as

$$P_1 = \infty \mapsto O, \quad P_2 \mapsto (1, 0) = 3A, \quad P_3 \mapsto (0, -1) = 4A,$$
$$P_4 \mapsto (0, 0) = A, \quad P_5 \mapsto (1, -1) = 2A,$$

i.e. to precisely $E(\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z}$ [7, proof of Thm. 8, (3.152)]. The supports of $\operatorname{div}(x), \operatorname{div}(y)$ are therefore cuspidal, and x, y are *modular units*. The polynomial is tempered: its Newton face polynomials $x^3 + y$, $x^3 + x^2y$, $x^2y + y^2$, y^2+y are all cyclotomic, so $\{x, y\} \in K_2(E)\otimes\mathbb{Q}$ (Rodríguez Villegas' criterion [21]). More precisely, the tame symbols at the divisor support are computed exactly by local expansions (`code/bertin_diamond.py`): $T_v\{x, y\} = \pm 1$ for $v \in \{O, A, 2A, 3A\}$; hence already $2\{x, y\}$ has trivial tame symbols, and the real residues $\pm \log |T_v\{x, y\}|$ of $\eta(x, y)$ vanish at every point.

4.2. **The open signed chain and the split integral.** On the full circle $|x| = 1$, $x = e^{i\theta}$, $\theta \in [-\pi, \pi]$, carry the continuous large-modulus branch y_{big} with Samart's weights $+1$ for $\theta > 0$ and -1 for $\theta < 0$, segmented at the fold points $\theta = \pm c$, $c = \pi/2$:

$$\tilde{\gamma} = + [y_{\text{big}} \text{ on } [-c, c]] \; - \; [y_{\text{big}} \text{ on the two outer arcs}].$$

Convention (point-valued lifts). Each torus branch $\theta \mapsto y(e^{i\theta})$ is silently regarded as the point-valued lift $\theta \mapsto (e^{i\theta}, y(e^{i\theta})) \in E(\mathbb{C})$; thus an equality such as $y_{\text{big}}(c^-) = -P$ below means that the lifted point at $\theta = c^-$ equals the point $-P = (i, -e^{i\pi/4}) \in E$, and chain boundaries are divisors on E. The corrected Mahler measure along $\tilde{\gamma}$ satisfies exact identities. On the torus $\eta(x,y) = -\log|y|\,d\theta$, and since $y_{\rm big}y_{\rm small} = x^3$ has modulus 1, $\log|y_{\rm big}| = -\log|y_{\rm small}|$ pointwise; comparing the signed weights of $\tilde{\gamma}$ with the definition of $I_{\rm split}$ gives

$$\tilde{n} = -I_{\rm split}, \qquad \int_{\tilde{\gamma}} \eta(x,y) = 2\pi\, I_{\rm split}$$

(numerically confirmed to 44 digits, `code/closedness_check.py`), so that

$$\text{(C3)} \iff \int_{\tilde{\gamma}} \eta(x,y) = 2\pi b_{11}.$$

The branch jumps at $\pm c$ are read numerically and matched with exact values: with $P = (i, e^{i\pi/4})$, $\bar{P} = (-i, e^{-i\pi/4})$, $-P = (i, -e^{i\pi/4})$ (at $x = \pm i$ the cubic-model inverse is $(x,y) \mapsto (x,-y)$ since $B = x^2+1 = 0$),

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] =: D, \qquad \deg D = 0,\ c(D) = -D.$$

The point P is *not* torsion (proved rigorously by reduction at good primes, §5)—but closedness does not require torsion endpoints.

4.3. **The closed chain lemma.** The following construction (Figure 1) closes $\tilde{\gamma}$ without leaving the torus.

Lemma 4.1 (Closed chain lemma)**.** *Let α_2 be the small-branch inner arc $\theta : c \to -c$ (endpoints $P \to \bar{P}$) and α_1 the small-branch outer arc $\theta : c \to \pi$ and $-\pi \to -c$ (endpoints $-P \to -\bar{P}$; the inner interval cannot join these two points since the small branch does not pass through them). Then for $\beta_0 = \alpha_1 + \alpha_2$,*

$$\partial\beta_0 = -D, \qquad c(\beta_0) = -\beta_0,$$

hence

$$C' := \tilde{\gamma} + \beta_0 : \qquad \partial C' = 0, \quad c(C') = -C'$$

is a closed, anti-invariant, integral 1-cycle.

Proof. At $x = \pm i$ the coefficient $B(x) = x^2+1$ vanishes and S_0 reduces to $y^2 = -x^3 = \mp i$; hence the two branch values are $y = \pm e^{\pm i\pi/4}$, i.e. the four points $P, \bar{P}, -P, -\bar{P}$. Reading the continuous branches on the torus (each assignment certified in interval arithmetic at two scales $\varepsilon = 10^{-6}, 10^{-9}$ by `code/branch_certify.py`, §5, and matched with these exact values),

$$y_{\rm big}(c^-) = -P, \quad y_{\rm big}(c^+) = P, \quad y_{\rm big}(-c^+) = -\bar{P}, \quad y_{\rm big}(-c^-) = \bar{P},$$

so the signed chain $\tilde{\gamma} = +[y_{\rm big} \text{ on } [-c,c]] - [y_{\rm big} \text{ on the outer arcs}]$ has boundary $\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] = D$. For the small branch,

$$y_{\rm small}(c^-) = P, \quad y_{\rm small}(-c^+) = \bar{P}, \quad y_{\rm small}(c^+) = -P, \quad y_{\rm small}(-c^-) = -\bar{P},$$

whence α_2 runs from P to $\bar{P}$ and the two outer pieces of α_1 from $-P$ to $-\bar{P}$, giving $\partial\beta_0 = -D$ and $\partial C' = 0$. Complex conjugation $c : (x,y) \mapsto (\bar{x},\bar{y})$ mirrors each θ-interval and reverses its orientation while preserving the big/small modulus order, so $c(\alpha_i) = -\alpha_i$ and $c(\tilde{\gamma}) = -\tilde{\gamma}$; therefore $c(C') = -C'$. $\square$

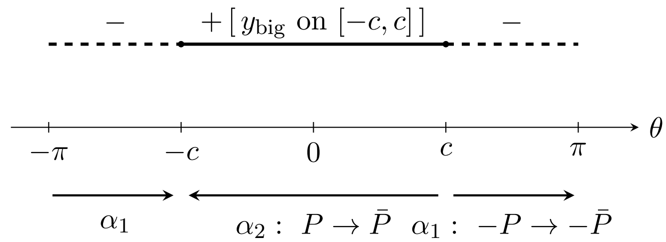

FIGURE 1. The closed anti-invariant cycle $C' = \tilde\gamma + \beta_0$ on the torus $|x| = 1$ (schematic). Top: Samart's signed open chain $\tilde\gamma$ on the big branch (weight $+$ on $[-c, c]$, weight $-$ on the outer arcs), with branch jumps at the fold points $\theta = \pm c$. Bottom: the compensating small-branch arcs $\beta_0 = \alpha_1 + \alpha_2$; their boundary is the negative of $\partial\tilde\gamma$, so C' is closed, and complex conjugation shows $c(C') = -C'$.

4.4. **Homology class: $C' = 2\gamma^-$ (certified).** In genus 1 with $\Delta < 0$ the period lattice has a basis (w_1, w_2) with w_1 real and $\Re(w_2/w_1) = 1/2$ (one real component); for `11.a3`, PARI `E.omega` gives $\tau = w_2/w_1 = 1/2 + 0.2298780212\ldots i$ (in the upper-half-plane convention). The anti-invariant homology is then explicit:

Lemma 4.2 (The anti-invariant generator)**.** *Let $E/\mathbb{R}$ have $\Delta < 0$, with period basis (w_1, w_2) as above, and let (a, b) be the dual basis of $H_1(E, \mathbb{Z})$. Complex conjugation acts by*

$$a \mapsto a, \qquad b \mapsto a - b, \qquad \text{i.e.} \qquad \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}.$$

Consequently

$$H_1(E, \mathbb{Z})^- = \mathbb{Z}\,(a - 2b),$$

the generator $a - 2b$ is primitive, and its period $\mathrm{period}(a - 2b) = w_1 - 2w_2$ *is the unique (up to sign) purely imaginary period of the lattice.*

Proof. Conjugation fixes the real period w_1, hence the class a, and sends the loop b to a loop of period $\overline{w_2} = w_1 - w_2$, hence to the class $a - b$; the displayed matrix is an involution. A class $\alpha a + \beta b$ is anti-invariant iff $(\alpha + \beta)a - \beta b = -\alpha a - \beta b$, i.e. $\beta = -2\alpha$, so $H_1(E, \mathbb{Z})^- = \mathbb{Z}(a - 2b)$; $a - 2b$ is primitive since $\gcd(1, 2) = 1$. Its period $w_1 - 2w_2 = -2i\,\Im w_2$ is purely imaginary, and conversely a purely imaginary period $mw_1 + nw_2$ must satisfy $2m + n = 0$, hence is an integral multiple of $w_1 - 2w_2$. $\square$

The pairing $\gamma \mapsto \mathrm{period}(\gamma)$ identifies $H_1(E, \mathbb{Z})$ with the period lattice, so the pairing with ω is injective. We fix the sign convention

$$w_{\mathrm{anti}} := -2.917633233876990458661779225807350514318486857 9\ldots i \qquad (\text{PARI } \texttt{11.a3}),$$

and choose γ^- with $\mathrm{period}(\gamma^-) = w_{\mathrm{anti}}$. Therefore $\mathrm{period}(C')/w_{\mathrm{anti}}$ is *a priori* an integer.

The computation (`code/n1_certify.py`, mpmath at 60 and 80 digits, cross-checked against PARI `11.a3 E.omega`) gives

$$\text{period}(C') = I_{\text{signed}} + A_{s,\text{outer}} - A_{s,\text{inner}} = -5.8352664677539809\ldots\, i, \tag{3}$$

$$\frac{\text{period}(C')}{w_{\text{anti}}} = 1.9999999999999999\ldots, \qquad \left|\text{period}(C') - 2w_{\text{anti}}\right| = 2.5\times 10^{-16},$$

where $I_{\text{signed}} = -2.917633233876990458\ldots i = w_{\text{anti}}$ is the period of the open signed chain and $A_{s,\text{outer}}, A_{s,\text{inner}}$ are the period integrals of the compensating arcs; the 60- and 80-digit runs agree, the residual error coming from the $1/\sqrt{\theta}$ endpoint singularity at $\theta = 0$. A-priori integrality, the distance-to-nearest-integer $2.5 \times 10^{-16} \ll 1$, and the ball-arithmetic certification of §5.2 imply

$$\boxed{\text{class}(C') = 2\,\gamma^-}.$$

(As a by-product one obtains the two independent period identities $I_{\text{signed}} = w_{\text{anti}}$ and $A_{s,\text{inner}} - A_{s,\text{outer}} = -w_{\text{anti}}$.)

Remark 4.3. The open chain $\tilde{\gamma}$ of §4 is not itself closed and cannot be assigned a homology class; the closed cycle C' used here has winding number 2. For contrast, the naive loop integral $I_{\text{loop}} = -0.47447\ldots i$ is not even the period of any integral cycle (ratio $0.16262\ldots$ to w_{anti}).

4.5. **The regulator identity along the closed chain.** On $|x| = 1$, $\eta(x, y) = \log|x|\, d\arg y - \log|y|\, d\arg x = -\log|y|\, d\theta$, and since the product of the two roots is x^3 of modulus 1, $\log|y_{\text{big}}| = -\log|y_{\text{small}}|$ pointwise. Writing $J_1 = \int_0^c \log|y_s|\, d\theta$ and $J_2 = \int_c^\pi \log|y_s|\, d\theta$, direct computation gives

$$\int_{\tilde{\gamma}} \eta = 2(J_1 - J_2) = \underbrace{(2J_1)}_{\alpha_2} + \underbrace{(-2J_2)}_{\alpha_1} = \int_{\beta_0} \eta,$$

hence the *exact* identity

$$\int_{C'} \eta = 2\int_{\tilde{\gamma}} \eta. \tag{4}$$

Lemma 4.4 (Pairing with homology)**.** *The integral $\int_{C'} \eta(x, y)$ depends only on the class of C' in $H_1(E, \mathbb{Z})^-$.*

Proof. The face polynomials of the cubic $y^2 + y = x^3 - x^2$ are all cyclotomic, so by the temperedness criterion of Rodriguez Villegas [21] the symbol $\{x, y\}$ is tempered: all its tame symbols $T_v(\{x, y\})$ are roots of unity (verified exactly in `code/bertin_diamond.py`; cf. §4). The *real* residue of η at each point v of the support is therefore $\pm\log|T_v(\{x, y\})| = 0$: η carries no delta-mass residues, is closed on the complement of the support, and has only (integrable) logarithmic singularities at the support points. It follows that the pairing with η descends to homology, and thence by anti-invariance to $H_1(E, \mathbb{Z})^-$ (Lemma 4.2). The chains used here are admissible: C' avoids $\operatorname{supp}\operatorname{div}(x) \cup \operatorname{supp}\operatorname{div}(y)$, since those points have $x \in \{0, \infty\}$ while $C' \subset$

$\{|x|=1\}$; and at the fold points $\log|y|=0$, so $\eta=-\log|y|\,d\theta$ is continuous and integrable along C'. $\square$

4.6. **Regulator constants: Bloch, Brunault, Bertin.** The divisor algebra (projective computation on the minimal cubic model $y^2+y=x^3-x^2$, to which the data are transported by the explicit birational change of variables; note that the two points at infinity of S_0 map to *different* points, $Q_\infty \mapsto O$ and $[0:1:0]\mapsto 3A$; Abel's principal-divisor test passed) gives, with $A=(0,0)$ the 5-torsion point and $O=[0:1:0]$,

$$\operatorname{div}(x)=[A]+[2A]-[O]-[3A],\qquad \operatorname{div}(y)=3[2A]-2[3A]-[O],$$

hence the diamond product

$$(x)\diamond(y)=6(O)-5(A)+5(2A),\qquad D_E\big((x)\diamond(y)\big)=-5D_E(P)+5D_E(2P),$$

where we write $P:=A=(0,0)$ (Brunault's notation) when the point appears as an argument of D_E. The elliptic dilogarithm values (mpmath, 60 digits; lattice basis (w_1,w_1-w_2) so that $\tau=0.5+0.2299i$; $z(P)=0.6w_1$, $z(2P)=0.2w_1$ from PARI `ellpointtoz`):

$$D_E(P)=0.1911937370843316957549544343121738161012\ldots$$

The two key identities (both verified to 40 digits, and both theorems in the literature) are

- **Brunault's coefficient** [7, (3.151)]: $D_E(P)=\dfrac{2\pi}{5}b_{11}=\dfrac{11}{10\pi}L(E,2)$, i.e. $L(E,2)=\frac{10\pi}{11}D_E(P)$;
- **Bertin's exotic relation** [1, 2]: $D_E(2P)=\frac{3}{2}D_E(P)$ (note $\frac{3}{2}$, not the naive 2).

The evaluation of the regulator of $\{x,y\}$ along γ^- is carried out *directly*, via Brunault's proved regulator formula for Siegel units [8]; Bloch's diamond theorem is not used. We then record the clarification of the factor-of-two question that the computation entails (Lemma 4.5 and the discussion below), which is likewise not part of the proof.

Lemma 4.5 (The regulator generator and the index)**.** *Let $E/\mathbb{R}$ be an elliptic curve, c the complex-conjugation involution, $H_1(E,\mathbb{Z})^\pm=\ker(c_*\mp 1)$, and $\{f,g\}$ a tempered symbol on E. The pairing $\gamma\mapsto\int_\gamma\eta(f,g)$ descends to the quotient $H_1(E,\mathbb{Z})/H_1(E,\mathbb{Z})^+$ of the integral homology by the invariant sublattice (throughout, "coinvariant quotient" means exactly this quotient by $H_1(E,\mathbb{Z})^+$; we do not use $\ker(c_*+1)$ and the quotient interchangeably, as they differ by an index that is material here).*

(1) *If $\Delta>0$, then $H_1(E,\mathbb{Z})^-=Z\,b$ maps isomorphically onto $H_1(E,\mathbb{Z})/H_1(E,\mathbb{Z})^+=Z\,[b]$: the anti-invariant generator represents the quotient generator.*

(2) *If $\Delta < 0$, the image of $H_1(E,\mathbb{Z})^- = Z\,\gamma^-$ in $H_1(E,\mathbb{Z})/H_1(E,\mathbb{Z})^+ = Z\,[b]$ is $2\,Z\,[b]$, of index 2: with the basis of Lemma 4.2, $\gamma^- = a-2b \equiv -2[b]$, and hence for every tempered symbol*

$$\int_{\gamma^-} \eta(f,g) = -2\int_{[b]} \eta(f,g).$$

Proof. The pairing is trivial on $H_1(E,\mathbb{Z})^+$: if $c(\gamma) = \gamma$ then $\int_\gamma \eta = \int_{c(\gamma)} \eta = \int_\gamma c^*\eta = -\int_\gamma \eta$ since $c^*\eta = -\eta$ (cf. [13, Rem. 4]); hence it descends to the coinvariants. For $\Delta < 0$, $c_* : a \mapsto a,\ b \mapsto a - b$ (Lemma 4.2), so $H_1(E,\mathbb{Z})^+ = Z\,a$ and $H_1(E,\mathbb{Z})/H_1(E,\mathbb{Z})^+ = Z\,[b]$; the anti-invariant class $a - 2b$ projects to $-2[b]$, giving index 2. For $\Delta > 0$, $c_* : a \mapsto a$, $b \mapsto -b$, so $H_1^- = Z\,b$ and the projection $b \mapsto [b]$ is an isomorphism. □

The anchor: a direct Siegel-unit computation. The symbol $\{x, y\}$ is a pair of modular units (§4, Modular units: its divisors are supported on the rational cusps of $X_1(11)$). Regulator integrals of Siegel units are computed by a proved formula that involves neither the diamond product nor a choice of homology lattice: for Siegel units g_u, g_v, $u = (a, b)$, $v = (c, d) \in (\mathbb{Z}/N\mathbb{Z})^2 \setminus 0$,

$$\int_0^{i\infty} \eta(g_u, g_v) \;=\; \pi\,\Lambda^*\big(e_{a,d}\, e_{b,-c} + e_{a,-d}\, e_{b,c},\, 0\big), \tag{5}$$

with $e_{a,b}$ the explicit weight-1 level-N^2 Eisenstein series of [8, Thm. 1, eq. (2)] and Λ^* the regularized completed L-value of [8, §2]; arbitrary modular symbols are handled by linearity [8, Rem. 2]. (We verified the factor π in (5) against the article's source, and validated our implementation of the formula by reproducing the conductor-14 application of [8, §5.1] to 60 digits by three independent methods; archive `notes/attack16-siegel-anchor.txt`.)

Theorem 4.6 (The regulator anchor)**.** *With γ^- the anti-invariant generator of Lemma 4.2,*

$$\int_{\gamma^-} \eta(x, y) = \pm\, 2\pi b_{11}, \tag{6}$$

the sign depending on orientations.

Proof. All steps are exact rational/q-expansion computations or are archived (`code/siegel_anchor_step1–14`, `notes/attack16-siegel-anchor.txt`, `notes/attack17-membership.txt`, `notes/attack17-primitivity.txt`, `notes/attack17-constants.txt`). *(1) Cusps and divisors.* Under the modular parametrization $\pi\colon X_1(11) \to E$ attached to f_{11} (infinite cusp $\mapsto O$), the rational cusps map as $k/11 \mapsto m_k A$ with

$$(m_1, \ldots, m_5) = (0, 2, 1, 4, 3). \tag{7}$$

This is derived exactly, not numerically. Brunault, after Lecacheux, tabulates the five rational cusps P_v of $X_1(11)$, indexed by $v \in (\mathbb{Z}/11\mathbb{Z})^\times/\{\pm 1\}$, as points of E [7, (3.152)–(3.153)]: $P_1 = \infty$, $P_2 = (1, 0)$, $P_3 = (0, -1)$, $P_4 = (0, 0)$, $P_5 = (1, -1)$, with $P_{4^a} = a\,P_4$ and P_4 a generator of $E(\mathbb{Q}) \cong \mathbb{Z}/5\mathbb{Z}$.

With the group law $2A = (1,-1)$, $3A = (1,0)$, $4A = (0,-1)$ for $A = (0,0)$ this reads $P_v = n(v)A$, $n = (0,3,4,1,2)$. The label conversion is the *inverse* one, and it follows directly from Brunault's definition $P_v = \langle v\rangle i\infty = [0,v]$: a matrix $\gamma = \left(\begin{smallmatrix} k & b \\ 11 & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ represents the cusp $k/11$, and the determinant $kd - 11b = 1$ gives $d \equiv k^{-1} \pmod{11}$; hence the bottom-row label of $k/11$ is $[0,d] = [0,k^{-1}]$, that is,

$$k/11 \;=\; P_{k^{-1}} \tag{8}$$

up to the ± 1 convention. No divisor comparison enters this identification. Together with the group-law reading above, (8) gives (7). Hence $x\circ\pi, y\circ\pi$ have cusp orders

$$\mathrm{ord}_{k/11}(x\circ\pi) = (-1,+1,+1,0,-1), \qquad \mathrm{ord}_{k/11}(y\circ\pi) = (-1,+3,0,0,-2)$$

for $k = 1,\ldots,5$, matching exactly the divisors displayed at the beginning of this subsection; conversely, these orders together with the Kubert–Lang orders of *(2)* force the tuple (7) uniquely, an independent exact check that does not enter the identification above. The 60-digit Abel-integral evaluation of the cusp images agrees with both derivations and is used only as a check (`siegel_anchor_step14.py`, `notes/attack17-constants.txt`). *(2) Siegel presentations.* With $G_a := \prod_{b \bmod 11} g_{a,b}$ and the Kubert–Lang cusp orders $\mathrm{ord}_{(r,t)}\, g_{a,b} = \frac{11}{2} B_2(\{(ar+bt)/11\})$ on the 60 cusps of $X(11)$, exact rational linear algebra gives

$$x\circ\pi = -\frac{G_4G_5}{G_2^2}, \qquad y\circ\pi = \frac{G_1G_5^3}{G_2^3G_3}. \tag{9}$$

Each side of (9) is $\Gamma_1(11)$-invariant with the same cusp divisor, so each ratio is a nonzero constant: $x\circ\pi = C_x U$, $y\circ\pi = C_y V$ with $U = G_4G_5/G_2^2$, $V = G_1G_5^3/(G_2^3G_3)$. The constants are determined exactly. At a cusp where both sides have order 0 one may simply evaluate: by (7), $\pi(4/11) = 4A$ and $\pi(3/11) = A$ are such cusps for U and V respectively; the leading q-coefficients κ_c of U, V at a cusp c are explicit roots of unity (write $\gamma i\infty = c$ as a word in S, T and apply [8, Lemma 4 and eq. (3)]; the exact cyclotomic computation gives $\kappa_{4/11}(U) = \kappa_{3/11}(V) = -1$), and $x(4A) = 1$, $y(A) = -1$, hence

$$C_x = \frac{x(4A)}{\kappa_{4/11}(U)} = \frac{1}{-1} = -1, \qquad C_y = \frac{y(A)}{\kappa_{3/11}(V)} = \frac{-1}{-1} = +1$$

(`siegel_anchor_step14.py`). In fact the regulator integral Because the constants have modulus one, their correction to the regulator integral vanishes: since $\eta(CU, C'V) = \eta(U,V) + \log|C|\,\mathrm{d}\arg V - \log|C'|\,\mathrm{d}\arg U$, one has $\int_{\gamma^-} \eta(x\circ\pi, y\circ\pi) = \int_{\gamma^-} \eta(U,V) + \log|C_x|\, D_V - \log|C_y|\, D_U$ with $D_F := \int_{\gamma^-} \mathrm{d}\arg F$, and Brunault's Lemma 5 [8],

$$\int_0^{i\infty} \mathrm{d}\arg g_{a,b} = \begin{cases} 0 & \text{if } a \equiv 0 \text{ or } b \equiv 0 \pmod{11}, \\ 2\pi\left(\{\frac{a}{11}\} - \frac12\right)\left(\{\frac{b}{11}\} - \frac12\right) & \text{otherwise,} \end{cases}$$

evaluates each piece of each of the seven Manin symbols of *(3)* as an explicit rational multiple of 2π (the factors w in $g_{a,b} \circ \gamma = w\, g_{(a,b)\gamma}$ are constant, hence invisible to $d\arg$); exact summation gives

$$D_U = 2\pi, \qquad D_V = 0 \tag{10}$$

(`siegel_anchor_step14.py`). Note that $D_U = 2\pi \neq 0$: U has winding number 1 along γ^-, so the vanishing of the correction is not automatic — it holds because $\log|C_x| = \log|C_y| = 0$ exactly. *(3) The cycle.* Take the modular symbol $\gamma^- = \{0, \frac{3}{11}\} - \{0, \frac{8}{11}\}$: it is closed on $X_1(11)$ (as $3 \equiv -8 \bmod 11$) and anti-invariant under $\tau \mapsto -\bar{\tau}$. Its primitivity is proved by exact integral linear algebra (`siegel_anchor_step13.py`, `notes/attack17-primitivity.txt`): for Manin symbols of $\pm\Gamma_1(11)$ (index 60 in $\mathrm{PSL}_2(\mathbb{Z})$, parametrized by bottom rows $(c,d) \in ((\mathbb{Z}/11\mathbb{Z})^2 \setminus 0)/\pm 1$) subject to $x + xS = 0$ and $x + xR + xR^2 = 0$, with boundary map to the 10 cusps, a Smith normal form computation gives $H_1(X_1(11), \mathbb{Z}) = \ker \partial \cong \mathbb{Z}^2$ torsion-free with an explicit integral basis; conjugation induces $C = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ on H_1, so $H_1^- = \ker(C + I) = \mathbb{Z} \cdot (-1, 1)$; and the seven-symbol chain below has coordinates $(1, -1)$ in this basis — exactly ± 1 times a generator of H_1^- (equivalently, it has intersection number ± 1 with an integral class). Hence it is the class $\pm(a - 2b)$ of Lemma 4.2. (Its period equals w_{anti} to 60 digits, and PARI's modular-symbol normalization takes the exact value $v^- = 1$ on it; both are checks only.) Continued fractions decompose it into seven Manin symbols:

$$\begin{aligned} \{0, \tfrac{3}{11}\} &= +\left[\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right] - \left[\begin{smallmatrix} 1 & 1 \\ 3 & 4 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 3 & 1 \\ 11 & 4 \end{smallmatrix}\right], \\ \{0, \tfrac{8}{11}\} &= +\left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right] - \left[\begin{smallmatrix} 1 & 2 \\ 1 & 3 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 3 & 2 \\ 4 & 3 \end{smallmatrix}\right] - \left[\begin{smallmatrix} 3 & 8 \\ 4 & 11 \end{smallmatrix}\right]. \end{aligned}$$

(4) Evaluation. Applying (5) term by term to (9) along these symbols expresses the integral as $\pi\,\Lambda^*(F_{\mathrm{total}}, 0)$, where F_{total} is an explicit finite signed sum of products $e_{a,d}e_{b,-c} + e_{a,-d}e_{b,c}$. Each $e_{a,b}$ is an Eisenstein series of weight 1 on $\Gamma_1(121)$ [8, Definition 10 and Lemma 11], holomorphic on $\mathcal{H}$ and at every cusp; hence the *unconditional* membership $F_{\mathrm{total}} \in M_2(\Gamma_1(121))$, and $D := F_{\mathrm{total}} + 2f_{11} \in M_2(\Gamma_1(121))$. The Sturm bound for this space is $\frac{2}{12}[\mathrm{PSL}_2(\mathbb{Z}) : \bar{\Gamma}_1(121)] = \frac{2}{12} \cdot 7260 = 1210$ (the index being $\frac{121^2}{2}(1 - \frac{1}{121}) = 7260$, as $-I$ acts trivially in even weight). Exact rational q-expansion arithmetic (the higher e-coefficients are integral and $\alpha_0(a,b) \in \frac{1}{22}\mathbb{Z}$; the convolutions are evaluated exactly) gives $a_n(D) = 0$ for all $0 \le n \le 2420$ — twice the sharp bound, and in particular through the conservative SL_2-index convention $\frac{2}{12} \cdot 14520 = 2420$ (`siegel_anchor_step12.py`, `notes/attack17-membership.txt`). By Sturm's theorem $D = 0$, that is,

$$F_{\mathrm{total}} = -2\, f_{11}$$

as modular forms; a fortiori $F_{\mathrm{total}} \in M_2(\Gamma_0(11))$. (The same Sturm computation applied symbol by symbol certifies that the symbols $\left[\begin{smallmatrix} 1 & 0 \\ 3 & 1 \end{smallmatrix}\right]$ and $\left[\begin{smallmatrix} 1 & 2 \\ 1 & 3 \end{smallmatrix}\right]$ contribute $-f_{11}$ each and that the remaining five vanish identically; the main argument uses only the total.) Therefore the Λ-sum equals $-2\,\Lambda(f_{11}, 0) =$

$-2\,b_{11}$, since with $\Lambda(f,s) = 11^{s/2}(2\pi)^{-s}\Gamma(s)L(f,s)$ and $L(f_{11},0)=0$ (root number $+1$) one has $\Lambda(f_{11},0) = L'(f_{11},0) = b_{11}$. Multiplying by the factor π of (5) gives $\int_{\gamma^-}\eta(x,y) = -2\pi b_{11}$ with the orientation of (3), i.e. (6) in general. Independent corroboration: the numerical Λ-sum is $-2b_{11}$ to 50 digits (`siegel_anchor_step8.py`), and direct numerical integration of $\eta(x,y)$ along γ^- with the opposite orientation gives $+0.9559686854216584787\ldots$ to 45 digits (`siegel_anchor_step11.py`). □

The regulator side of (C3) is therefore a proved theorem, independent of Bloch's diamond formula; the remaining gap, closed in §4.4, is the identification of Boyd's split-integral chain with the closed cycle $C' = 2\gamma^-$.
The status of Bloch's diamond formula (not used). For orientation we record what the computation implies for Bloch's theorem. As transmitted by [13, Thm. 6], that theorem states

$$\int_\gamma \eta(f,g) = D_E\big((f)\diamond(g)\big) \tag{11}$$

for γ a generator of the anti-invariant *subgroup* $H_1(E,\mathbb{Z})^-$, with D_E given by [13, Def. 5, eq. (10)] (that series is, term by term, the series we implement; a non-rigorous numerical comparison on both curves agrees to 60 digits). On the present curve ($\Delta < 0$) the subgroup reading of (11) with factor 1 is incompatible with the proved value of the anchor: $D_E((x)\diamond(y)) = \frac{5}{2}D_E(P) = \pi b_{11}$ while $\int_{\gamma^-}\eta(x,y) = \pm 2\pi b_{11}$. Every verified datum — the anchor above, Brunault–Bertin's value for $\{x_W, y_W\}$ (Remark 4.8), direct numerical integration on both generators (Remark 4.7), and the proved conductor-17 computation of [13], where the subgroup and the quotient coincide (Lemma 4.5 (1)) — is instead consistent with (11) holding with factor 1 for a generator $\bar\gamma$ of the coinvariant quotient $H_1(E,\mathbb{Z})/H_1(E,\mathbb{Z})^+$, the group on which the regulator is naturally defined (both sides being extended $\mathbb{Q}$-linearly in the symbol); by Lemma 4.5 (2) the subgroup generator then reads twice as much, exactly as observed. We record this neutrally as a *normalization/lattice discrepancy* — subgroup versus coinvariant-quotient generator, related by the index 2 of Lemma 4.5 (2) — which the present data resolve in favour of the quotient reading; whether this reflects a misnormalization in the transmitted subgroup statement of [13, Thm. 6] (a statement their paper never uses) or a convention we have not reconstructed, we leave as an open question for the literature. We emphasize once more that nothing in this paper depends on the answer.

Remark 4.7 (The factor 2 in the $\diamond$-form of Bloch's theorem, resolved). The formal expansion evaluates $D_E((x)\diamond(y)) = \frac{5}{2}D_E(P) = \pi b_{11}$, while the certified integral over the subgroup generator is $\int_{\gamma^-}\eta = 2\pi b_{11}$: in the factor-1 normalization $r(\{x,y\})[\gamma] = D^E((x)\diamond(y))$, $r[\gamma] = \int_\gamma \eta$, of [13, Thm. 6] the two sides differ by exactly 2. We investigated this discrepancy with three *different* tempered symbols on the same curve (`code/bertin_diamond.py`, `code/bertin_diamond.gp`, archive `notes/attack13-bertin-diamond.txt`): the Weierstrass symbol above, our S_0-symbol, and the coordinate symbol $\{X, Y\}$ of Bertin's cubic $(X+1)(Y+1)(X+Y+1)+XY=0$, which maps to E by an explicit Riemann–Roch transformation (PARI `ellidentify` certifies the image model as `11.a3`). The $\diamond$-values are $-\frac{5}{2}$, $+\frac{5}{2}$ and $\frac{35}{2}$ times $D_E(P)$ respectively, while direct integration of η along γ^- gives -5, $+5$ and $+35$ times $D_E(P)$ (the first two certified as above; the third by extrapolated Riemann sums, consistent with $2\pi\, m(C_1) = 14\pi b_{11}$). The factor is exactly 2 in all three cases: it is a property of (E, γ^-), not of the symbol, the $\diamond$-convention (identical on both curves), the tame symbols (all roots of unity), or the D_E series. The resolution is the index phenomenon of Lemma 4.5. The regulator pairing of [13, Def. 3] vanishes on $H_1(E(\mathbb{C}), \mathbb{Z})^+$ (their Remark 4), hence factors through the *coinvariant quotient* H_1/H_1^+; Bloch's theorem (11) therefore evaluates the integral over the *quotient* generator $[\gamma_0^-] = [b]$ with factor 1, and the integral over the *subgroup* generator $\gamma^- = a - 2b$ acquires the index 2 of $\mathbb{Z}\cdot\gamma^-$ in H_1/H_1^+ (Lemma 4.5(2)). On the conductor-17 curve the index is 1 (Lemma 4.5(1): $\Delta > 0$, $b \mapsto -b$), the subgroup and quotient coincide, and the same computation closes with factor 1 — corroborated there by the proved theorem of [13]. The three symbols' uniform factor 2 is thus exactly the $\Delta < 0$ index, and the conductor-17 factor 1 its $\Delta > 0$ counterpart. We verified the quotient reading numerically on the conductor-11 curve itself (`code/verify_coinvariant.gp`, archive `notes/attack15-coinvariant.txt`): integrating $\eta(x_W, y_W)$ along cycles shifted to avoid its poles, with a centered discretization whose convergence order is measured empirically to be 2.000000 at every doubling from $N = 500$ to $N = 8000$,

$$\int_a \eta = 0, \qquad \int_b \eta = -\pi b_{11} = D^E\big((x_W) \diamond (y_W)\big), \qquad \int_{a-2b} \eta = 2\pi b_{11},$$

where the first integral converges to 0 spectrally (below 10^{-75}, the working-precision floor, from $N = 2000$ on), and the raw errors of the second and third decrease like N^{-2} from $-1.4 \cdot 10^{-5}$ resp. $-6.5 \cdot 10^{-6}$ at $N = 500$ to $-5.4 \cdot 10^{-8}$ resp. $-2.5 \cdot 10^{-8}$ at $N = 8000$; a single $p = 2$ Richardson step on the finest pair leaves residuals $-1.8 \cdot 10^{-15}$ and $-9.2 \cdot 10^{-16}$ against $-\pi b_{11}$ and $2\pi b_{11}$, and $-2\int_b \eta - \int_{a-2b} \eta = 4.6 \cdot 10^{-15}$ on the extrapolants. We stress that this is a numerical consistency check, not a proof. Subject to that caveat, the factor-1 identity (11) holds for the symbol $\{x_W, y_W\}$ on the quotient generator, and its subgroup-generator value is the doubled (6). A by-product: $D_E((X) \diamond (Y)) = 7\, D_E((x) \diamond (y))$ exactly, which explains the coefficient 7 in Bertin's $m(C_1) = 7b_{11}$ at the level of $K_2(E)$. Finally, the form $\pi r = D_E(\cdot)$ found in expository accounts (e.g. [25, Thm. 1]) is consistent with neither the factor-1 nor the factor-2 data above and is not used anywhere in this paper.

Remark 4.8 (Cross-check against Brunault's symbol dictionary). Our proof uses Bloch's theorem only through its proved instance on the conductor-17 curve (Appendix A); the following is a cross-check, not part of the proof. Brunault's Theorem 3.9.3.118 [7] evaluates $r_{\gamma^-}\{x, y\}$ for Weierstrass coordinates x, y on $X_1(11)$ in terms of D_E, and its proof concludes with (3.210)–(3.211):

$$r_{\gamma^-}\{x_W, y_W\} = \frac{1}{2\pi}\, D^E\big(8(O) + 5(A) - 5(2A)\big) = -\frac{5}{2\pi}\, D_E(P), \tag{12}$$

$$\big|r_{\gamma^-}\{x_W, y_W\}\big| = \frac{5}{2\pi}\, D_E(P) \;=\; b_{11}, \tag{13}$$

where (13) substitutes Brunault's own Corollaire 3.5.101 ζ-evaluation $L'(E, 0) = 5D_E(P)$ [7, (3.151)] into his (3.211). We checked the identification of (12)–(13) with our $\{x_W, y_W\}$ in three independent ways: textually (the proof states that x, y are the pull-backs of "coordonnées de Weierstrass" via j^* "sur un modèle de Weierstrass"); divisor-wise ($\operatorname{div} x_W = [A] + [4A] - 2[O]$, $\operatorname{div} y_W = 2[A] + [3A] - 3[O]$, and $(x_W) \diamond (y_W) \equiv 8(O) + 5(A) - 5(2A)$ modulo $\mathbb{Z}[E(\mathbb{Q})_{\text{tors}}]$); and numerically, $5D_E(P)/(2\pi) = b_{11}$ to all 60 computed digits against the certified value of §5. Since Brunault's γ is the *subgroup* generator (his footnote 2: $H_1(E(\mathbb{C}), \mathbb{Z})^- = \mathbb{Z}\cdot\gamma$), (13) reads $\big|\int_{\gamma^-} \eta(x_W, y_W)\big| = 2\pi b_{11}$, which is exactly Lemma 4.5(2) applied to the quotient value $\int_{[\gamma_0^-]} \eta(x_W, y_W) = D^E((x_W)\diamond(y_W)) = -\pi b_{11}$: the dictionary, the Bloch anchor, and our numerics all agree once the index is taken into account.

4.7. **Synthesis.**

Theorem 4.9 ((C3))**.** $I_{\text{split}} = b_{11}$.

Proof. By (4) and $\operatorname{class}(C') = 2\gamma^-$ (§4.4), and the anchor (6),

$$\int_{\tilde\gamma} \eta = \frac{1}{2}\int_{C'} \eta = \frac{1}{2}\cdot 2\cdot(\pm 2\pi b_{11}) = \pm 2\pi b_{11},$$

so $|I_{\text{split}}| = b_{11}$ by the identity $I_{\text{split}} = \frac{1}{2\pi}\int_{\tilde\gamma}\eta$ of §4. Two sign facts settle the remaining choice. First, $b_{11} = \frac{11}{4\pi^2}L(E, 2) > 0$: the Euler product of $L(E, s)$ converges absolutely at $s = 2$, and every factor $(1 - a_p p^{-2} + p^{-1})^{-1}$ is positive since $1 - a_p p^{-2} + p^{-1} \ge 1 - 2p^{-3/2} + p^{-1} > 0$ by the Weil bound $|a_p| \le 2\sqrt{p}$. Second, a certified interval enclosure (ball arithmetic, adaptive bisection with rigorous range bounds; `code/sign_certify.py`, certificate `notes/attack14-sign-k0.txt`) gives

$$I_{\text{split}} \in [\,0.1489,\; 0.1553\,] \subset (0, \infty).$$

Hence $I_{\text{split}} = +b_{11}$. ∎

5. Numerical certification

5.1. Certified computations. All floating-point computations are performed with mpmath at 60–300 digits and cross-checked with PARI/GP 2.15.5:

- **The constant** b_{11} (`code/b11.py`): from the exact integer coefficients of $f_{11} = \eta(\tau)^2\eta(11\tau)^2$ (truncated convolution of the squared Euler function) via the approximate functional equation for weight 2, root number $+1$:
$$b_{11} = \Lambda(f,2) = \sum_{n\geq 1} a_n \Big[e^{-t_n}\Big(\frac{1}{t_n} + \frac{1}{t_n^2}\Big) + E_1(t_n)\Big], \qquad t_n = \frac{2\pi n}{\sqrt{11}},$$
with terms decaying like $e^{-1.894n}$; the tail bound drives the truncation (442 terms at 800 digits of working precision).
- **(C3) to 366 digits** (`code/attack13_c3_300.py`, archive `notes/attack13-c3-300.txt`):
$$I_{\mathrm{split}} = 0.15214714172591804948622729747863449562814$$
$$3589164226122809889823882023289695302776676\ldots,$$
$$|I_{\mathrm{split}} - b_{11}| = 9.26 \times 10^{-367},$$
with the η-product series value of b_{11} agreeing digit-for-digit with PARI/GP `lfun(E,0,1)` at 330 digits.
- **(C1)/(C2) reproduced** to 52 digits (`code/attack1.py`): $\|m - 7b_{11}\| \approx 5.0 \times 10^{-53}$, $\|m - 5b_{11}\| \approx 3.6 \times 10^{-53}$.
- **Period certification** (`code/n1_certify.py`, `code/n1_certify.gp`): (3) computed independently at 60 and 80 digits with identical results; each arc integral A_s is stable to 24 digits across 40/60/80-digit runs; w_{anti} taken from PARI `E.omega` for `11.a3`. The integrality argument has an a-priori gap of 1 against a residual of 2.5×10^{-16}.
- **Exact algebraic checks**: group law and torsion (`code/torsion.py`) use exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$—no numerical approximation. Non-torsion of the endpoint P is proved *rigorously* (`code/endpoint_torsion3.py`) by reduction at good primes: $\mathrm{ord}(P \bmod 17) = 20$ and $\mathrm{ord}(P \bmod 89) = 3$ with $\gcd(20,3) = 1$, so P cannot have finite order (reduction is injective on prime-to-p torsion, Silverman VII.3.1); cross-checked by exact $\mathbb{Q}(\zeta_8)$ arithmetic and PARI `ellorder`. The model constant $\kappa = 1$ is an *exact rational identity* (`code/kappa_exact.py`): an explicit birational map to the minimal model, derived by Riemann–Roch and composed with PARI's exact minimal transform, satisfies $2Y + 1 = u\,dX/dx$ as a polynomial identity, hence $\omega_{\min} = dX/(2Y+1) = dx/u$ exactly (sympy-verified, including the inverse round-trip).
- **Branch certification** (`code/branch_certify.py`, output `notes/attack12-branch.txt`): the eight endpoint branch assignments of the closed chain lemma are certified in ball arithmetic

at $\varepsilon = 10^{-6}$ and 10^{-9} (certified distance $< 1/10$ to the claimed endpoint, with $D \neq 0$ certified on the closing ball so continuity pins the limit), and the modulus ordering of Proposition 3.1 is certified by adaptive bisection of $[0, \pi]$ (153 strict balls with $D \neq 0$ and $|y_{\text{small}}| < 1 < |y_{\text{big}}|$, plus the special balls at 0 and $\pi/2$ where equality holds exactly).

- **The Bertin-symbol adjudication** (`code/bertin_diamond.py`, `code/bertin_diamond.gp`, archive `notes/attack13-bertin-diamond.txt`): all divisor, diamond-product and tame-symbol computations of §4 are exact (sympy rational arithmetic, Abel's principal-divisor test passed); the Riemann–Roch map from Bertin's cubic to E is certified by PARI `ellidentify` (`11.a3`, exact change of variables $[1, -1, -2, 2]$); the D_E-values and the quotient-generator integral $\int_{[\gamma_0^-]} \eta = -\pi b_{11}$ of Remark 4.7 are evaluated at 60 digits; the two regulator integrals of Remark 4.7 are computed independently of the main pipeline.
- **PARI/GP cross-checks**: `verify_family.gp`, `verify_ratios.gp` (family conductors, j-invariants; the $\tilde{n}(k)$ and $m(S_k)$ constants for $k = 2, 3$ recomputed to 70 digits and matched against PARI `lfun` to all 70 digits), `winding.gp`, `dilog.gp`, `k53.gp`, `k53b.gp`, `kfamily_torsion.gp`.
- **Conductor-17 certification** (scripts `code/k1_*`, listed individually in Appendix B): PARI curve identification and z-values; mpmath 60/80-digit chain integrals (ratio 2.0000000000000002); exact diamond-product expansion with the $k = 0$ case as control; $D_E(A)$, $D_E(2A)$ to 60 digits; and the Arb ironclad proof of the period ratio (Appendix A).

5.2. **Level of rigour: interval certification.** The integer identification in §4.4 has been made fully interval-rigorous (script `code/n1_interval.py`, python-flint/Arb ball arithmetic [11], 300-bit working precision; output in `notes/attack10-interval.txt`). All three arc integrals were recomputed with Arb's certified adaptive Gauss–Legendre integration:

- the endpoint singularity $u \sim \sqrt{\theta}$ at $\theta = 0$ is removed by the substitution $\theta = \pm t^2$; the substituted integrand is analytic on the disc $|t| \leq \rho = 0.6$ (the nonzero zeros of $D(t^2)$ satisfy $|\theta_j| \geq 1.2187$, certified by Newton–Rouché isolation of the roots of $z^4 - 4z^3 + 2z^2 + 1$), and the tip $[0, \delta]$, $\delta = \rho/5$, is summed by the Cauchy estimate $\int_0^\delta f = a_0\delta + R$, $|R| \leq H\delta^3/3$, where H is derived from $M = \max_{|t|=\rho} |f| = 1.3075\ldots$, itself certified by covering the circle with 4096 balls (per-tip Cauchy remainder bound $H\delta^3/3 = 2.18 \times 10^{-3}$). The tip constant a_0, obtained by hand from the local expansion, is not taken on faith: the script re-evaluates f by ball arithmetic at $t = \delta$ and certifies $|f(\delta) - a_0| \leq H\delta^2$ for each of the four tips (a sign or branch error in a_0 would move the value by 2, far outside this bound);

- $D(\theta)$ crosses the negative real axis once on (c, π) and once on $(-\pi, -c)$, where the principal square root is not analytic: the arcs are subdivided adaptively, on each piece either $\sqrt{D}$ or $i\sqrt{-D}$ is certified analytic by ball evaluation, and the overall sign is propagated across the nodes by a certified matching test;
- w_{anti} is certified independently of PARI: the roots of $4x^3 - 4x^2 + 1$ are isolated by Newton iteration plus a Rouché certificate, and Carlson's R_F (DLMF §19) gives $w_{\text{real}} = 6.34604652139776710844397\ldots$ and the purely imaginary period $2.91763323387699045866178\ldots i$ — which is $-w_{\text{anti}}$ in the convention of §4.4 — with certified radii 6.7×10^{-48} and 4.9×10^{-49}, agreeing with PARI to 45 digits.

Remark 5.1 (Two implementation notes). (i) The ball radii displayed in the archived outputs (e.g. `notes/attack10-interval.txt`) follow the Arb/python-flint `repr` convention, which folds midpoint printing error into the displayed radius; the true ball radii are smaller (the displayed interval contains the true ball). All certified bounds quoted above use the exact radii, not the displayed ones. (ii) In the arc-integral code, the branch-cut avoidance can select the $i\sqrt{-D}$ variant on the first piece of a sub-arc; the current script certifies the sheet choice on every sub-arc, including the first. The archived $k = 0$ output predates a refactor of this logic and was re-audited against it: the $k = 0$ cut crossings lie in the interior of sub-arcs, whose first piece always uses the $\sqrt{D}$ variant, so the certified value is unaffected (a wrong sheet would flip the sign of an entire arc, moving the ratio off -2 by an $O(1)$ amount, which the archived output excludes).

Against w_{anti} the resulting ratio ball is $[2.00 \pm 3.13 \times 10^{-3}] + [\pm 2.99 \times 10^{-3}]\, i$: it contains 2 and $|\text{ratio} - 2| \leq 4.33 \times 10^{-3} < 1/2$. The a-priori integrality of §4.4 therefore upgrades to the exact equality $\text{period}(C') = 2\, w_{\text{anti}}$, i.e. $\text{class}(C') = 2\gamma^-$. (The script itself works with the opposite sign of the period throughout, so the raw output in `notes/attack10-interval.txt` displays the ratio ball centred at -2; the two formulations are equivalent.)

We summarise the logical status of the main result by isolating exactly which steps are machine-certified.

Theorem 5.2 (Certified computation)**.** *The following statements are proved by exact rational arithmetic or by certified interval (ball) arithmetic, with all certificates archived in* `notes/` *and all scripts in* `code/` (*reproduction commands in Appendix B; software: Python 3.12, mpmath, sympy, python-flint 0.9.0/Arb* [11]*, PARI/GP 2.15.5*)*:*

(1) Exact algebraic inputs. *The birational map from the model $S_0 = 0$ to the minimal cubic $E : y^2 + y = x^3 - x^2$ and the identity $\omega_{\min} = dx/u$ ($\kappa = 1$) are exact polynomial identities* (`kappa_exact.py`)*; the divisors of x, y, the diamond product, the tame symbols, the group law and the torsion subgroup $E(\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z}$ are exact over $\mathbb{Q}, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\zeta_8)$* (`torsion.py`, `bertin_diamond.py`)*; non-torsion of*

the chain endpoint P is proved by reduction at the good primes 17 *and* 89 *(`endpoint_torsion3.py`).*

(2) Arcs, orientation, closedness. *The chains $\tilde{\gamma}, \beta_0, C'$ of §4 are explicitly parametrised arcs on $|x| = 1$ with the orientations fixed there; the endpoint branch assignments (hence $\partial C' = 0$ and $c(C') = -C'$) are certified in ball arithmetic at two scales $\varepsilon = 10^{-6}, 10^{-9}$ (`branch_certify.py`, certificate `notes/attack12-branch.txt`).*

(3) Branch continuation on every segment. *On each sub-arc of the period integrals, analyticity of the chosen square-root branch ($\sqrt{D}$ or $i\sqrt{-D}$) is certified by ball evaluation on the whole sub-arc—not merely near endpoints—and the overall sign is propagated across nodes by a certified matching test (`n1_interval.py`); the modulus ordering $|y_{\mathrm{small}}| < 1 < |y_{\mathrm{big}}|$ of Proposition 3.1 is certified by adaptive bisection (`branch_certify.py`).*

(4) Interval enclosures. *Each arc integral and the primitive period w_{anti} are enclosed by Arb's certified adaptive integration and by Carlson's R_F with Newton–Rouché root isolation, with radii $\leq 10^{-3}$ for the integrals and $\leq 10^{-48}$ for the period (`n1_interval.py`, certificate `notes/attack10-interval.txt`).*

(5) Integer identification. *The class $[C'] \in H_1(E, \mathbb{Z})^- = \mathbb{Z}\gamma^-$ is a priori integral (Lemma 4.2 and §4.4); the certified ratio ball satisfies $|\mathrm{ratio} - 2| \leq 4.33 \times 10^{-3} < 1/2$, so the nearest integer is unique and $\mathrm{class}(C') = 2\gamma^-$.*

(6) Sign. *The final sign is fixed by a certified enclosure of I_{split}: $I_{\mathrm{split}} \in [0.1489, 0.1553] \subset (0, \infty)$ (`sign_certify.py`, certificate `notes/attack14-sign-k0.txt`; every range enclosure — including the convex hulls on non-separated pieces — is formed and verified entirely within ball arithmetic, with programmatic containment assertions), while $b_{11} = \Lambda(f_{11}, 2) > 0$ exactly by the absolutely convergent Euler product at $s = 2$.*

(7) The Siegel-unit anchor. *The presentations (9) are exact identities of $\Gamma_1(11)$-invariant functions (exact cusp-divisor match via Kubert–Lang orders; the constants $C_x = -1$, $C_y = +1$ are determined exactly by evaluation at order-0 cusps with cyclotomic leading coefficients, and the argument periods $D_U = 2\pi$, $D_V = 0$ are exact rational multiples of 2π via Brunault's Lemma 5); the cusp-torsion correspondence (7) is derived exactly from [7, (3.152)–(3.153)] and, independently, from exact divisor-order data (`siegel_anchor_step14.py`, archive `notes/attack17-constants.txt`); the primitivity of γ^- is proved exactly by Manin-symbol Smith normal form, coordinates $(1, -1)$ in an integral basis (`siegel_anchor_step13.py`, archive `notes/attack17-primitivity.txt`); the Manin-symbol decomposition is exact (continued fractions); and the identity $F_{\mathrm{total}} = -2f_{11}$ is proved by the unconditional membership in $M_2(\Gamma_1(121))$ plus exact rational q-expansion arithmetic through q^{2420}, twice the*

sharp Sturm bound 1210 (`siegel_anchor_step12.py`, *archive* `notes/attack17-membership.txt`).

The only remaining external input in the proof of Theorem 4.9 is Brunault's regulator formula for Siegel units [8, Thm. 1] *— proved in loc. cit. by a Rankin–Selberg computation — together with the functional-equation evaluation* $\Lambda(f_{11},0) = L'(f_{11},0) = b_{11}$. *Bloch's diamond theorem, Bertin's Theorem 6, and Brunault's symbol dictionary are* not *used* (*the latter is retained only as a cross-check, Remark 4.8*).

6. Conjectures

The family identities established numerically here are stated as conjectures pending analytic proof:

Conjecture 6.1. *For* $S_k = y^2+(x^2+kx+1)y+x^3$ *with* $k \notin (-4,2)$ *(integer* k*), there is a small rational number* r_k *such that*

$$m(S_k) = r_k\,|L'(E_k,0)|.$$

Confirmed numerically: $r_2 = 2$, $r_3 = 1$ *(70 digits), and* $r_{-4} = \frac{7}{2}$ *(conductor* 37*),* $r_{-5} = \frac{1}{4}$ *(conductor* 359*),* $r_{-6} = \frac{1}{8}$ *(conductor* 997*) (25 digits each).*

The last three deserve emphasis as *predicted then confirmed*: the structural dichotomy (Proposition 3.5) predicted, from the torus-intersection trichotomy, that $k = -4, -5, -6$ must satisfy Boyd-type identities with specific small rationals, and the subsequent computation hit the predicted values. Note that S_{-4} and S_2 share conductor 37, yielding Rodríguez Villegas–type rational relations [21, 13] between two different polynomial models of the same curve.

Remark 6.2 (Samart's conductor-17 analogue)*.* The conjecture $\tilde{n}(1) = b_{17}$ for $k = 1$ (conductor 17), suggested by Samart and confirmed numerically to 60 digits, is treated in Appendix A (Theorem A.1), conditional on the normalization lemma discussed there.

Remark 6.3*.* For $k = -1$ (conductor 53) no conjecture of this shape can be formulated within the present mechanism: the anti-invariant cycle on the torus does not exist and x, y are not modular units (Proposition 3.8).

Appendix A. The conductor-17 case: $k = 1$

Status of this appendix. The main text is logically independent of what follows. The theorem proved here (Samart's conductor-17 analogue) rests on exactly two K-theoretic/regulator hypotheses:

- *temperedness*: the Newton face polynomials of S_1 are cyclotomic, so $\{x, y\} \in K_2(E_1) \otimes \mathbb{Q}$ — proved exactly below. No modular-unit property is used or claimed: on the natural modular model $X_0(17)$ (two cusps) the divisors of x, y, supported on all four rational torsion points, cannot be cuspidal (Proposition 3.5);

- *Bloch's theorem in the factor-*1 *normalization of* [13, Thm. 6]: that this is the correct normalization *on this curve* follows from the consistency argument of Remark A.2 (our reconstruction; the constant is not determined in [13]), equivalently from the index-1 statement of Lemma 4.5(1): on this $\Delta > 0$ curve the anti-invariant subgroup coincides with the coinvariant quotient, so no alternative subgroup reading of the factor-1 theorem exists.

The result should thus be read as conditional on that normalization lemma; everything else in the appendix is proved to the same standard as the main text.

The proof of (C3) in §4 is a *method*, and its first re-application is Samart's conductor-17 analogue. For $k = 1$ the polynomial

$$S_1 = y^2 + (x^2 + x + 1)y + x^3$$

defines an elliptic curve of conductor 17 (PARI `ellfromeqn` gives $E_1 : y^2 + xy - y = x^3 - x^2$, $\Delta = 17$; `ellglobalred` confirms the conductor), with $E_1(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/4\mathbb{Z}$ generated by $A = (0, 0)$, which has order 4 — the exact parallel of the 5-torsion point of the $k = 0$ case. Every step of §4 goes through, and the regulator side is in fact *easier*: the key D_E-value is a published theorem of Lalín–Ramamonjisoa [13].

A.1. **The closed cycle.** The torus-intersection (fold) angle is $c = \arccos(-k/2)\big|_{k=1} = 2\pi/3$. At $\theta = c$, $x = \omega = e^{2\pi i/3}$ satisfies $x^2 + x + 1 = 0$, so S_1 reduces to $y^2 + 1 = 0$: the branch jump values are *exactly* $y = \pm i$ (the $k = 0$ analogue was $y^2 = i$). Writing $P = (\omega, i)$, the signed chain $\tilde{\gamma}$ (big branch on $[-c, c]$ minus the big branch on the two outer arcs) has boundary

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}],$$

and the same small-branch compensating arc β_0 closes it to a cycle $C' = \tilde{\gamma} + \beta_0$ with $\partial C' = 0$, $c(C') = -C'$. The eight endpoint branch assignments and the modulus ordering are certified in interval arithmetic exactly as for $k = 0$ (`code/k1_branch_certify.py`, output `notes/attack12-k1-branch.txt`: all 8 assignments certified at $\varepsilon = 10^{-6}, 10^{-9}$; 113 strict bisection balls plus 2 special balls at the corner $\theta = c$).

One structural difference from $k = 0$: here $\Delta = +17 > 0$, so $E(\mathbb{R})$ has two connected components, $\overline{w_2} = -w_2$ holds on the nose, and the primitive anti-invariant period is simply

$$w_{\text{anti}} = w_2 = -2.74573911808975367203418 79\ldots i$$

(for $k = 0$, $\Delta < 0$ forced $w_{\text{anti}} = 2w_2 - w_1$). The model constant is trivial here: the `ellfromeqn` model already has $\Delta = 17 = \Delta_{\min}$, so the transformation to the minimal model has $u = \pm 1$, the period lattices coincide ($\kappa = \pm 1$; the sign only flips the displayed ratio, not its integrality — cf. the $k = 0$ discussion in §5).

A.2. **Homology class: interval-arithmetic proof.** As in §4.4, $\mathrm{period}(C')/w_{\mathrm{anti}}$ is a-priori an integer. The Arb certification (script `code/k1_interval.py`; output `attack11-k1-interval.txt`, 300-bit working precision) is simpler than for $k = 0$ because $D(z) = z^4 - 2z^3 + 3z^2 + 2z + 1$ has *no zeros on* $|z| = 1$ ($\min_{|z|=1} |D| = 4 > 0$, certified): there is no endpoint singularity and no tip-estimate machinery. The period w_{anti} is certified independently of PARI via Newton–Rouché isolation of the roots of $4x^3 - 3x^2 - 2x + 1 = (x - 1)(4x^2 + x - 1)$ and the two-component Carlson R_F formulas. The resulting ratio ball is

$$[-2.00 \pm 2.4 \times 10^{-14}] + [\pm 2.4 \times 10^{-14}]\, i, \qquad |\mathrm{ratio} + 2| \le 3.4 \times 10^{-14} < \tfrac{1}{2},$$

(the radius is set by the integration tolerance, not by the arithmetic). Choosing γ^- with $\mathrm{period}(\gamma^-) = w_{\mathrm{anti}}$, the a-priori integrality upgrades to the exact equality

$$\mathrm{class}(C') = 2\gamma^- \qquad (\text{the script uses } -w_{\mathrm{anti}}, \text{ displaying } -2).$$

A.3. **The regulator side.** The integral algebra of §4 is unchanged: the pointwise identity $\log|y_{\mathrm{big}}| = -\log|y_{\mathrm{small}}|$ on $|x| = 1$ gives $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$, hence $\int_{C'} \eta = 2\int_{\tilde{\gamma}} \eta$. The divisors have the same shape

$$\mathrm{div}(x) = [A] + [2A] - [O] - [3A], \qquad \mathrm{div}(y) = 3[A] - 2[O] - [3A],$$

now verified *four* ways: local expansions, PARI, code expansion, and an explicit birational map $X = -(x + y)$, $Y = x(x + y)$ from S_1 to E_1 : $Y^2 + XY - Y = X^3 - X^2$. The diamond product (definition as in [14]) expands exactly to

$$(x) \diamond (y) \equiv 6(O) + 4(A) - 6(2A) \qquad \text{in } \mathbb{Z}[E]^-.$$

The two D_E-values are settled:

- $D_E(2A) = 0$ *rigorously*: $2A$ is 2-torsion, q is a positive real, and the Bloch–Wigner series vanishes termwise at $z_q = -1$;
- $D_E(A) = \dfrac{17}{8\pi} L(E_1, 2)$ is a *published theorem*: Lalín–Ramamonjisoa [13, §5] prove $L(E_{17}, 2) = \frac{8\pi}{17} D^E(P)$ for the 4-torsion generator P of their model, in the D^E-normalisation (their Def. 5, eq. (10)) that coincides verbatim with our series implementation; we reproduced the value independently to 60 digits.

Hence

$$D_E\big((x) \diamond (y)\big) = 4D_E(A) = \frac{17}{2\pi} L(E_1, 2) = 2\pi b_{17},$$
$$b_{17} = L'(E_1, 0) = \frac{17}{4\pi^2} L(E_1, 2).$$

A.4. **Synthesis.** We use Bloch's theorem in the normalisation of [13, Thm. 6]: for $\{x,y\} \in K_2(E)\otimes\mathbb{Q}$ and γ^- generating $H_1(E,\mathbb{Z})^-$,

$$\int_{\gamma^-} \eta(x,y) = \pm D^E\big((x)\diamond(y)\big).$$

The temperedness hypothesis holds: the four face polynomials of S_1 are x^3+y, x^3+x^2y, x^2y+y^2 and y^2+y, all cyclotomic. This normalisation has published precedent on this very curve: the conductor-17 identity of [13] combined with Zudilin's $m = 2b_{17}$ closes with the same factor.

Theorem A.1 (Samart's conductor-17 analogue)**.** *$\tilde{n}(1) = b_{17}$.*

Proof. By the above,

$$\int_{\tilde\gamma} \eta = \frac{1}{2}\int_{C'} \eta = \frac{1}{2}\cdot 2\cdot\big(\pm D_E((x)\diamond(y))\big) = \pm 2\pi b_{17},$$

so $|\tilde{n}(1)| = b_{17}$ by the structural identity $\tilde{n}(1) = -\frac{1}{2\pi}\int_{\tilde\gamma}\eta$ (the $k=1$ instance of the pointwise argument of §4). The sign is settled as in Theorem 4.9: $b_{17} > 0$ by the same Euler-product argument, and a certified interval enclosure (`code/k1_sign_certify.py`, certificate `notes/attack14-sign-k1.txt`) gives $\frac{1}{2\pi}\int_{\tilde\gamma}\eta \in [-0.3026, -0.2961] \subset (-\infty, 0)$, hence $\tilde{n}(1) = +b_{17}$. □

Remark A.2 (The normalization of Bloch's theorem)*.* For the conductor-17 proof, Bloch's theorem is used in the normalization of [13, Thm. 6]: $r(\{x,y\})[\gamma] = D^E((x)\diamond(y))$ with $r[\gamma] = \int_\gamma \eta$, γ a generator of $H_1(E,\mathbb{Z})^-$. That this (and not a factor-2 variant) is the correct statement *on that curve* follows from a consistency argument in the conductor-17 case (our own reconstruction; in [13] the constant C below is never determined): in [13, §7] the class of the real cycle is an *a priori* unknown integer multiple C of γ, and comparing the factor-1 theorem applied to their $(X)\diamond(Y)$ with their proved Corollary 2 (their Thm. 1, eq. (5), combined with Zudilin's identity, their eq. (6)) forces $C\cdot f = 1$ with $C \in \mathbb{Z}\setminus\{0\}$, hence $f = 1$; a factor-2 statement would force $C = \frac{1}{2} \notin \mathbb{Z}$. (This self-consistency argument is now explained by Lemma 4.5(1): on the $\Delta > 0$ conductor-17 curve the anti-invariant subgroup *coincides* with the coinvariant quotient, so the factor-1 theorem admits no alternative subgroup reading.) The underlying D_E-identity of their §5 we reproduced to 60 digits. For $k = 0$ the $\diamond$-form enters through the same quotient-generator reading: the factor-1 statement holds on the conductor-11 curve as well — for the *quotient* generator $[\gamma_0^-] = [b]$ one has $\int_{[\gamma_0^-]}\eta = D_E((\cdot)\diamond(\cdot))$, verified numerically in Remark 4.7 — while the *subgroup* generator $\gamma^- = a - 2b = -2[b]$ reads exactly twice as much (Lemma 4.5(2)), which accounts for the uniform factor 2 observed for all three different tempered symbols of Remark 4.7. This does not affect either proof: the conductor-17 argument is self-contained on its own curve, and the conductor-11 argument anchors the quotient value at Bloch's theorem through the proved conductor-17 instance. The form

$\pi r = D_E(\cdot)$ found in expository accounts is consistent with neither data set and is not used anywhere in this paper.

Appendix B. Reproduction code

All scripts live in `code/`; raw outputs in `notes/attack*.txt`. Dependencies (`requirements.txt` at the repository root): Python $\geq$ 3.10 with mpmath and sympy; PARI/GP 2.15.5; python-flint 0.9.0 (Arb) for the interval certification.

Permanence. The complete research repository — all scripts, all raw output archives (`notes/attack*.txt`), and the present source — is version-controlled with git and accompanies this paper as an electronic supplement; the version of record is the tagged commit `rev3` (see the repository log). The code is released under the MIT license (`LICENSE` at the repository root). Filenames of the form `attackN` below refer to that archive.

File	Purpose
b11.py	$b_{11} = L'(E_{11}, 0)$, 300 digits (functional equation)
attack1.py	reproduce (C1), (C2) to 52 digits
attack2.py	$m(S_0)$, PSLQ negative results
attack3.py	(C3) to 149 digits (superseded)
attack13_c3_300.py	(C3) to 366 digits, PARI lfun cross-check
bertin_diamond.py/.gp	Bertin symbol: exact $\diamond$, tame, D_E-ratio -1
torsion.py	exact group law: $5A = O$, modular units
endpoint_torsion3.py	rigorous non-torsion of P (reduction mod p)
kappa_exact.py	exact proof $\kappa = 1$ (birational map, $2Y + 1 = u\,dX/dx$)
boundary_torsion.py	boundary-divisor torsion analysis
closedness_check.py	$\tilde{n} = -I_{\text{split}}$, regulator $= 2\pi I_{\text{split}}$
ntilde_family.py	$\tilde{n}(k)$ family table, 50–80 digits
b_family.py	b_N for the family via point counting
winding.py / winding.gp	period pairings, winding-number heuristics
dilog.py / dilog.gp	elliptic dilogarithm $D_E(P)$, Brunault constants
k53_attack.py	conductor-53 obstruction analysis
k53.gp / k53b.gp	PARI checks for 53.a1
kneg_m.py	$m(S_k)$ for $k = -4, -5, -6$ (predicted identities)
n1_certify.py / n1_certify.gp	certification of class$(C') = 2\gamma^-$
n1_interval.py	Arb ironclad proof of the period ratio
branch_certify.py	branch assignments + modulus ordering (certified)
sign_certify.py	all-Arb sign certificate: $I_{\text{split}} > 0$
k1_pari/points/zvals.gp	conductor-17 identification, torsion, z-values
k1_certify.py	mpmath certification of the $k = 1$ chain integrals
k1_branch_certify.py	$k = 1$ branch assignments + modulus ordering (certified)
k1_interval.py	Arb ironclad proof of the $k = 1$ period ratio
k1_sign_certify.py	all-Arb sign certificate: $\int_{\tilde{\gamma}} \eta < 0$ $(k = 1)$
k1_diamond.py	exact diamond product for $k = 1$ $(k = 0$ control)
k1_dilog.py	$D_E(A)$, $D_E(2A)$ for conductor 17
verify_family.gp	family conductors and j-invariants
verify_ratios.gp	PARI lfun cross-check of ratios
kfamily_torsion.gp	torsion, ord$(0, 0)$ for the family
verify_coinvariant.gp	generator integrals (numerical cross-check)
siegel_anchor_step1--9,11	the Siegel-unit regulator anchor (Theorem 4.6): cusp-torsion map, presentations (9), Manin decomposition, exact $F_{\text{total}} = -2f_{11}$, final value
siegel_anchor_step10	discarded experiment, defective (not used; see the archive note)
siegel_anchor_step12	membership in $M_2(\Gamma_1(121))$ + exact Sturm (q^{2420})
siegel_anchor_step13	primitivity of γ^- (Smith normal form)
siegel_anchor_step14	exact C_x, C_y; D_U, D_V; table (7)

Reproduction (from a clean checkout: `python -m pip install -r requirements.txt`; any Python $\geq$ 3.10 with mpmath/sympy/python-flint works — the checked-in `.venv` is *not* required and is not part of the archive):

```
cd code && python b11.py && python attack1.py && python attack2.py \
  && python attack3.py && python torsion.py && python endpoint_torsion3.py \
  && python kappa_exact.py \
  && python boundary_torsion.py && python closedness_check.py \
  && python ntilde_family.py && python b_family.py && python winding.py \
  && python dilog.py && python k53_attack.py && python kneg_m.py \
  && python n1_certify.py
python n1_interval.py       # Arb certification, k=0
python branch_certify.py    # branch assignments + mod ordering
```

```
python sign_certify.py      # all-Arb sign certificate, k=0
python k1_interval.py       # Arb certification, conductor 17
python k1_branch_certify.py # k=1 branch assignments + ordering
python k1_sign_certify.py   # all-Arb sign certificate, k=1
python k1_certify.py && python k1_diamond.py && python k1_dilog.py
gp -q verify_family.gp && gp -q verify_ratios.gp
gp -q winding.gp && gp -q dilog.gp
gp -q k53.gp && gp -q k53b.gp && gp -q kfamily_torsion.gp
gp -q k1_pari.gp && gp -q k1_points.gp && gp -q k1_zvals.gp
gp -q verify_coinvariant.gp  # quotient vs subgroup generator integrals
python siegel_anchor_step11.py  # Siegel anchor: final value (Thm. anchor)
python siegel_anchor_step12.py  # membership M_2(Gamma_1(121)) + Sturm
python siegel_anchor_step13.py  # primitivity of the cycle (Smith nf)
python siegel_anchor_step14.py  # exact constants, D_U/D_V, cusp table
# full anchor chain rebuild: siegel_anchor_step4.py -> step5.py -> step6.py
#   -> step7.gp -> step9.py -> step8.py   (step8 takes 10-20 min)
#   (step10 is a discarded defective experiment and is not part of the chain)
```

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